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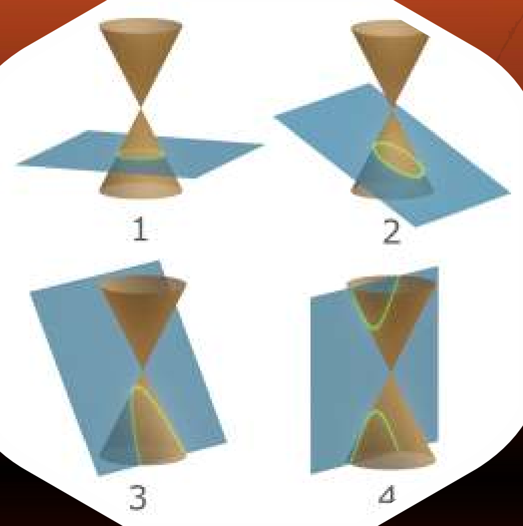
# MATHEMATICS

## JEE (MAIN+ADVANCED)

NURTURE COURSE




Study Material

## Conic Section

(English Medium)

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# PARABOLA

## 1. CONIC SECTIONS :

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.

**(a)** The fixed point is called the **focus**.

**(b)** The fixed straight line is called the **directrix**.

**(c)** The constant ratio is called the **eccentricity** denoted by e.

**(d)** The line passing through the focus & perpendicular to the directrix is called the **axis**.

**(e)** A point of intersection of a conic with its axis is called a **vertex**.

## 2. GENERAL EQUATION OF A CONIC : FOCAL DIRECTRIX PROPERTY :

The general equation of a conic with focus $(p, q)$ & directrix $lx + my + n = 0$ is :

$$(l^2 + m^2)\,[(x - p)^2 + (y - q)^2] = e^2\,(lx + my + n)^2 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

## 3. DISTINGUISHING BETWEEN THE CONIC :

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e. Two different cases arise.

**Case (i) When the focus lies on the directrix :**

In this case $D \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ & the general equation of a conic represents a pair of straight lines and if :

$e > 1$ the lines will be real & distinct intersecting at S.

$e = 1$ the lines will coincident.

$e < 1$ the lines will be imaginary.

**Case (ii) When the focus does not lie on the directrix :**

The conic represents:

| a parabola | an ellipse | a hyperbola | a rectangular hyperbola |
|---|---|---|---|
| $e = 1$ ; $D \neq 0$<br>$h^2 = ab$ | $0 < e < 1$ ; $D \neq 0$<br>$h^2 < ab$ | $D \neq 0$ ; $e > 1$ ;<br>$h^2 > ab$ | $e > 1$ ; $D \neq 0$<br>$h^2 > ab$ ; $a + b = 0$ |

## 4. PARABOLA :

A parabola is the locus of a point which moves in a plane, such that its distance from a fixed point (focus) is equal to its perpendicular distance from a fixed straight line (directrix).

Standard equation of a parabola is $y^2 = 4ax$. For this parabola :

(i) Vertex is $(0, 0)$ (ii) Focus is $(a, 0)$ (iii) Axis is $y = 0$ (iv) Directrix is $x + a = 0$

**(a) Focal distance :**

The distance of a point on the parabola from the focus is called the **focal distance of the point.**

**(b) Focal chord :**

A chord of the parabola, which passes through the focus is called a **focal chord**.

**(c) Double ordinate :**

A chord of the parabola perpendicular to the axis of the symmetry is called a **double ordinate**.

**(d) Latus rectum :**

A double ordinate passing through the focus or a focal chord perpendicular to the axis of parabola is called the **latus rectum**. For $y^2 = 4ax$.

- Length of the latus rectum = **4a.**
- Length of the semi latus rectum = **2a**.
- Ends of the latus rectum are **L(a, 2a ) & L'(a, – 2a )**

**Note that :**

(i) Perpendicular distance from focus on directrix = half the latus rectum.

(ii) Vertex is middle point of the focus & the point of intersection of directrix & axis.

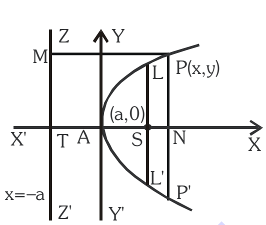
(iii) Two parabolas are said to be equal if they have the same latus rectum.

## 5. PARAMETRIC REPRESENTATION :

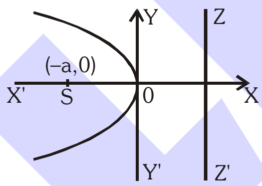
The simplest & the best form of representing the co-ordinates of a point on the parabola is $(at^2, 2at)$ . The equation $x = at^2$ & $y = 2at$ together represents the parabola $y^2 = 4ax$ , t being the parameter.

## 6. TYPE OF PARABOLA :

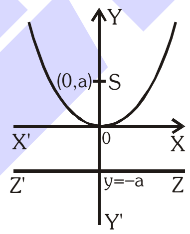
Four standard forms of the parabola are $y^2 = 4ax$ ; $y^2 = -4ax$ ; $x^2 = 4ay$ ; $x^2 = -4ay$



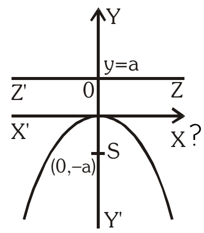

$y^2 = 4ax$




$y^2 = -4ax$




$x^2 = 4ay$




$x^2 = -4ay$

| Parabola | Vertex | Focus | Axis | Directrix | Length of Latus rectum | Ends of Latus rectum | Parametric equation | Focal length |
|---|---|---|---|---|---|---|---|---|
| $y^2 = 4ax$ | (0,0) | (a,0) | y=0 | x=–a | 4a | (a, ±2a) | $(at^2, 2at)$ | x + a |
| $y^2 = -4ax$ | (0,0) | (–a,0) | y=0 | x=a | 4a | (–a, ±2a) | $(-at^2, 2at)$ | x–a |
| $x^2 = +4ay$ | (0,0) | (0,a) | x=0 | y=–a | 4a | (±2a, a) | $(2at, at^2)$ | y+a |
| $x^2 = -4ay$ | (0,0) | (0,–a) | x=0 | y=a | 4a | (±2a, –a) | $(2at, -at^2)$ | y–a |
| $(y-k)^2 = 4a(x-h)$ | (h,k) | (h+a,k) | y=k | x+a–h=0 | 4a | (h+a, k±2a) | $(h+at^2, k+2at)$ | x–h+a |
| $(x-p)^2 = 4b(y-q)$ | (p,q) | (p, b+q) | x=p | y+b–q=0 | 4b | (p±2a,q+a) | $(p+2at, q+at^2)$ | y–q+b |

***Illustration 1 :*** Find the vertex, axis, directrix, focus, latus rectum and the tangent at vertex for the parabola $9y^2 - 16x - 12y - 57 = 0$.

***Solution :*** The given equation can be rewritten as $\left(y-\frac{2}{3}\right)^2 = \frac{16}{9}\left(x+\frac{61}{16}\right)$ which is of the form $Y^2 = 4AX$. Hence the vertex is $\left(-\frac{61}{16}, \frac{2}{3}\right)$

The axis is $y-\frac{2}{3}=0 \Rightarrow y=\frac{2}{3}$

The directrix is $X+A=0 \Rightarrow x+\frac{61}{16}+\frac{4}{9}=0 \Rightarrow x=-\frac{613}{144}$

The focus is $X = A$ and $Y = 0 \Rightarrow x+\frac{61}{16}=\frac{4}{9}$ and $y-\frac{2}{3}=0$

$\Rightarrow \quad \text{focus}=\left(-\frac{485}{144},\frac{2}{3}\right)$

Length of the latus rectum $=4A=\frac{16}{9}$

The tangent at the vertex is $X=0 \Rightarrow x=-\frac{61}{16}$. **Ans.**

***Illustration 2 :*** The length of latus rectum of a parabola, whose focus is (2, 3) and directrix is the line $x - 4y + 3 = 0$ is -

(A) $\frac{7}{\sqrt{17}}$ (B) $\frac{14}{\sqrt{21}}$ (C) $\frac{7}{\sqrt{21}}$ (D) $\frac{14}{\sqrt{17}}$

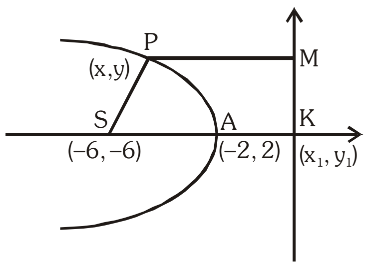
***Solution :*** The length of latus rectum $= 2 \times$ perp. from focus to the directrix

$$=2\times\left|\frac{2-4(3)+3}{\sqrt{(1)^2+(4)^2}}\right|=\frac{14}{\sqrt{17}}$$ **Ans. (D)**

***Illustration 3 :*** Find the equation of the parabola whose focus is (–6, –6) and vertex (–2, 2).

***Solution :*** Let S(–6, –6) be the focus and A(–2, 2) is vertex of the parabola. On SA take a point $K(x_1, y_1)$ such that SA = AK. Draw KM perpendicular on SK. Then KM is the directrix of the parabola. Since A bisects SK, $\left(\frac{-6+x_1}{2},\frac{-6+y_1}{2}\right)=(-2,2)$

$\Rightarrow \quad -6 + x_1 = -4$ and $-6 + y_1 = 4$ or $(x_1, y_1) = (2, 10)$




Hence the equation of the directrix KM is

$y - 10 = m(x - 2)$ ......... (i)

Also gradient of SK $=\frac{10-(-6)}{2-(-6)}=\frac{16}{8}=2; \Rightarrow m=\frac{-1}{2}$

$y-10=\frac{-1}{2}(x-2)$ (from (i))

$\Rightarrow \quad x + 2y - 22 = 0$ is the directrix

Next, let PM be a perpendicular on the directrix KM from any point P(x, y) on the parabola. From SP = PM, the equation of the parabola is $\sqrt{\left\{(x+6)^2+(y+6)^2\right\}}=\frac{|x+2y-22|}{\sqrt{(1^2+2^2)}}$

or $\quad 5(x^2 + y^2 + 12x + 12y + 72) = (x + 2y - 22)^2$

or $\quad 4x^2 + y^2 - 4xy + 104x + 148y - 124 = 0$

or $\quad (2x - y)^2 + 104x + 148y - 124 = 0$. **Ans.**

***Illustration 4 :*** The extreme points of the latus rectum of a parabola are (7, 5) and (7, 3). Find the equation of the parabola.

***Solution :*** Focus of the parabola is the mid-point of the latus rectum.

$\Rightarrow$ S is (7, 4). Also axis of the parabola is perpendicular to the latus rectum and passes through the focus. Its equation is

$$y - 4 = \frac{0}{5-3}(x-7) \Rightarrow y = 4$$

Length of the latus rectum = (5 – 3) = 2

Hence the vertex of the parabola is at a distance 2/4 = 0.5 from the focus. We have two parabolas, one concave rightwards and the other concave leftwards.

The vertex of the first parabola is (6.5, 4) and its equation is $(y-4)^2 = 2(x-6.5)$ and it meets the x-axis at (14.5, 0). The equation of the second parabola is $(y-4)^2 = -2(x-7.5)$. It meets the x-axis at (–0.5, 0). **Ans.**

**Do yourself - 1 :**

**1.** Name the conic represented by the equation $\sqrt{ax} + \sqrt{by} = 1$, where a, b $\in$ R, a, b, > 0.

**2.** Find the vertex, axis, focus, directrix, latus rectum of the parabola $4y^2 + 12x - 20y + 67 = 0$.

**3.** Find the equation of the parabola whose focus is (1, –1) and whose vertex is (2, 1). Also find its axis and latus rectum.

**4.** Find the equation of the parabola whose latus rectum is 4 units, axis is the line $3x + 4y = 4$ and the tangent at the vertex is the line $4x - 3y + 7 = 0$.

**5.** The equation of the directrix of the parabola, $y^2 + 4y + 4x + 2 = 0$ is -

(A) $x = -1$ (B) $x = 1$ (C) $x = -3/2$ (D) $x = 3/2$

**6.** Length of the latus rectum of the parabola $25[(x-2)^2 + (y-3)^2] = (3x - 4y + 7)^2$ is-

(A) 4 (B) 2 (C) 1/5 (D) 2/5

Find the vertex, axis, latus rectum, and focus of the parabolas :

**7.** $y^2 = 4x + 4y$ **8.** $x^2 + 2y = 8x - 7$ **9.** $x^2 - 2ax + 2ay = 0$ **10.** $y^2 = 4y - 4x$

**11.** Prove that the equation to the parabola, whose vertex and focus are on the axis of x at distances a and a' from the origin respectively, is $y^2 = 4(a' - a)(x - a)$.

**12.** PQ is a double ordinate of a parabola. Find the locus of its point of trisection.

**13.** A double ordinate of the curve $y^2 = 4px$ is of length 8p ; prove that the lines from the vertex to its two ends are at right angles.

**14.** 'O' is the vertex of the parabola $y^2 = 4ax$ & L is the upper end of the latus rectum. If LH is drawn perpendicular to OL meeting OX in H, prove that the length of the double ordinate through H is $4a\sqrt{5}$.

**15.** A point P on a parabola $y^2 = 4x$, the foot of the perpendicular from it upon the directrix, and the focus are the vertices of an equilateral triangle, find the area of the equilateral triangle.

## 7. POSITION OF A POINT RELATIVE TO A PARABOLA :

The point $(\mathbf{x_1}, \mathbf{y_1})$ lies **outside**, **on** or **inside** the parabola $\mathbf{y^2 = 4ax}$ according as the expression $\mathbf{y_1^2 - 4ax_1}$ is **positive**, **zero** or **negative**.

***Illustration 5 :*** Find the value of $\alpha$ for which the point $(\alpha - 1, \alpha)$ lies inside the parabola $y^2 = 4x$.

***Solution :*** $\because$ Point $(\alpha - 1, \alpha)$ lies inside the parabola $y^2 = 4x$

$\therefore \quad y_1^2 - 4ax_1 < 0$

$\Rightarrow \quad \alpha^2 - 4(\alpha - 1) < 0$

$\Rightarrow \quad \alpha^2 - 4\alpha + 4 < 0$

$(\alpha - 2)^2 < 0 \Rightarrow \alpha \in \phi$ **Ans.**

## 8. CHORD JOINING TWO POINTS :

The equation of a chord of the parabola $y^2 = 4ax$ joining its two points $P(t_1)$ and $Q(t_2)$ is

$\mathbf{y(t_1 + t_2) = 2x + 2at_1t_2}$

**Note :**

(i) If PQ is focal chord then $\mathbf{t_1t_2 = -1}$.

(ii) Extremities of focal chord can be taken as $\mathbf{(at^2, 2at)}$ & $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$

***Illustration 6 :*** Through the vertex O of a parabola $y^2 = 4x$ chords OP and OQ are drawn at right angles to one another. Show that for all position of P, PQ cuts the axis of the parabola at a fixed point.

***Solution :*** The given parabola is $y^2 = 4x$ ...... (i)

Let $P \equiv (t_1^2, 2t_1)$, $Q \equiv (t_2^2, 2t_2)$

Slope of OP $= \frac{2t_1}{t_1^2} = \frac{2}{t_1}$ and slope of OQ $= \frac{2}{t_2}$

Since OP $\perp$ OQ, $\frac{4}{t_1t_2} = -1$ or $t_1t_2 = -4$ ...... (ii)

The equation of PQ is $y(t_1 + t_2) = 2(x + t_1t_2)$

$\Rightarrow \quad y\left(t_1 - \frac{4}{t_1}\right) = 2(x - 4)$ [from (ii)]

$\Rightarrow \quad 2(x - 4) - y\left(t_1 - \frac{4}{t_1}\right) = 0 \Rightarrow \quad L_1 + \lambda L_2 = 0$

$\therefore$ variable line PQ passes through a fixed point which is point of intersection of $L_1 = 0$ & $L_2 = 0$

i.e. (4, 0) **Ans.**

## 9. LINE & A PARABOLA :

**(a)** The line y = mx + c meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as a >=< cm ⇒ condition of tangency is, $\mathbf{c = \frac{a}{m}}$.

**Note :** Line y = mx + c will be tangent to parabola $x^2 = 4ay$ if $\mathbf{c = -am^2}$.

**(b)** Length of the chord intercepted by the parabola $y^2 = 4ax$ on the line y = mx + c is : $\left(\frac{4}{m^2}\right)\sqrt{a(1+m^2)(a-mc)}$.

**Note :** Length of the focal chord making an angle α with the x - axis is $4a\,\text{cosec}^2\,\alpha$.

***Illustration 7 :*** If the line $y = 3x + \lambda$ intersect the parabola $y^2 = 4x$ at two distinct points then set of values of λ is -

(A) $(3, \infty)$ (B) $(-\infty, 1/3)$ (C) $(1/3, 3)$ (D) none of these

***Solution :*** Putting value of y from the line in the parabola -

$(3x + \lambda)^2 = 4x$

$\Rightarrow \quad 9x^2 + (6\lambda - 4)x + \lambda^2 = 0$

$\because \quad$ line cuts the parabola at two distinct points

$\therefore \quad D > 0$

$\Rightarrow \quad 4(3\lambda - 2)^2 - 4.9\lambda^2 > 0$

$\Rightarrow \quad 9\lambda^2 - 12\lambda + 4 - 9\lambda^2 > 0$

$\Rightarrow \quad \lambda < 1/3$

Hence, $\lambda \in (-\infty, 1/3)$ **Ans.(B)**

**Do yourself - 2 :**

**1.** Find the value of 'a' for which the point $(a^2 - 1, a)$ lies inside the parabola $y^2 = 8x$.

**2.** The focal distance of a point on the parabola $(x - 1)^2 = 16(y - 4)$ is 8. Find the co-ordinates.

**3.** Show that the focal chord of parabola $y^2 = 4ax$ makes an angle α with x-axis is of length $4a\,\text{cosec}^2\alpha$.

**4.** Find the condition that the straight line ax + by + c = 0 touches the parabola $y^2 = 4kx$.

**5.** Find the length of the chord of the parabola $y^2 = 8x$, whose equation is x + y = 1.

**6.** Prove that the locus of the middle points of all chords of the parabola $y^2 = 4ax$ which are drawn through the vertex is the parabola $y^2 = 2ax$.

**7.** In the parabola $y^2=6x$, find (1) the equation to the chord through the vertex and the negative end of the latus rectum, and (2) the equation to any chord through the point on the curve whose abscissa is 24.

**8.** Locus of the feet of the perpendiculars drawn from vertex of the parabola $y^2 = 4ax$ upon all such chords of the parabola which subtend a right angle at the vertex is

(A) $x^2 + y^2 - 4ax = 0$ (B) $x^2 + y^2 - 2ax = 0$

(C) $x^2 + y^2 + 2ax = 0$ (D) $x^2 + y^2 + 4ax = 0$

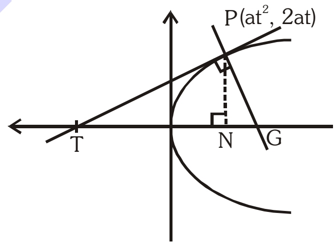
**9.** The triangle PQR of area 'A' is inscribed in the parabola $y^2 = 4ax$ such that the vertex P lies at the vertex of the parabola and the base QR is a focal chord. The modulus of the difference of the ordinates of the point Q and R is :

(A) $\frac{A}{2a}$ (B) $\frac{A}{a}$ (C) $\frac{2A}{a}$ (D) $\frac{4A}{a}$

**10.** Through the focus of the parabola $y^2 = 2px (p > 0)$ a line is drawn which intersects the curve at $A(x_1,y_1)$ and $B(x_2,y_2)$. The ratio $\frac{y_1y_2}{x_1x_2}$ equals-

(A) 2 (B) –1 (C) –4 (D) some function of p

## 10. LENGTH OF SUBTANGENT & SUBNORMAL :




PT and PG are the tangent and normal respectively at the point P to the parabola $y^2 = 4ax$. Then

TN = length of subtangent = twice the abscissa of the point P

(Subtangent is always bisected by the vertex)

NG = length of subnormal which is constant for all points on the parabola & equal to its semilatus rectum (2a).

## 11. TANGENT TO THE PARABOLA $y^2 = 4ax$ :

**(a) Point form :**

Equation of tangent to the given parabola at its point $(x_1, y_1)$ is

$$\mathbf{yy_1 = 2a\,(x + x_1)}$$

**(b) Slope form :**

Equation of tangent to the given parabola whose slope is 'm', is

$$y = mx + \frac{a}{m}, (m \neq 0)$$

Point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

**(c) Parametric form :**

Equation of tangent to the given parabola at its point P(t), is $\mathbf{ty = x + at^2}$

**Note :** Point of intersection of the tangents at the point $t_1$ & $t_2$ is $[at_1t_2, a(t_1+ t_2)]$.

***Illustration 8 :*** A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. Find its equation and its point of contact.

***Solution :*** Let the slope of the tangent be m

$$\therefore \quad \tan 45° = \left|\frac{3-m}{1+3m}\right| \Rightarrow \quad 1+3m = \pm(3-m)$$

$$\therefore \quad m = -2 \quad \text{or} \quad \frac{1}{2}$$

As we know that equation of tangent of slope m to the parabola $y^2 = 4ax$ is $y = mx + \frac{a}{m}$

and point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

for $m = -2$, equation of tangent is $y = -2x - 1$ and point of contact is $\left(\frac{1}{2}, -2\right)$

for $m = \frac{1}{2}$, equation of tangent is $y = \frac{1}{2}x + 4$ and point of contact is (8, 8) **Ans.**

***Illustration 9 :*** Find the equation of the tangents to the parabola $y^2 = 9x$ which go through the point (4, 10).

***Solution :*** Equation of tangent to parabola $y^2 = 9x$ is

$$y = mx + \frac{9}{4m}$$

Since it passes through (4, 10)

$$\therefore \quad 10 = 4m + \frac{9}{4m} \quad \Rightarrow \quad 16m^2 - 40\,m + 9 = 0$$

$$m = \frac{1}{4}, \frac{9}{4}$$

$\therefore$ equation of tangent's are $y = \frac{x}{4} + 9$ & $y = \frac{9}{4}x + 1$ **Ans.**

***Illustration 10 :*** Find the locus of the point P from which tangents are drawn to the parabola $y^2 = 4ax$ having slopes $m_1$ and $m_2$ such that -

(i) $m_1^2 + m_2^2 = \lambda$ (constant) (ii) $\theta_1 - \theta_2 = \theta_0$ (constant)

where $\theta_1$ and $\theta_2$ are the inclinations of the tangents from positive x-axis.

***Solution :*** Equation of tangent to $y^2 = 4ax$ is $y = mx + a/m$

Let it passes through P(h, k)

$$\therefore \quad m^2h - mk + a = 0$$

(i) $m_1^2 + m_2^2 = \lambda$

$$(m_1 + m_2)^2 - 2m_1m_2 = \lambda$$

$$\frac{k^2}{h^2} - 2.\frac{a}{h} = \lambda$$

$\therefore$ locus of P(h, k) is $y^2 - 2ax = \lambda x^2$

(ii) $\theta_1 - \theta_2 = \theta_0$

$\tan(\theta_1 - \theta_2) = \tan\theta_0$

$$\frac{m_1 - m_2}{1 + m_1 m_2} = \tan\theta_0$$

$$(m_1 + m_2)^2 - 4m_1 m_2 = \tan^2\theta_0 (1 + m_1 m_2)^2$$

$$\frac{k^2}{h^2} - \frac{4a}{h} = \tan^2\theta_0 \left(1 + \frac{a}{h}\right)^2$$

$$k^2 - 4ah = (h + a)^2 \tan^2\theta_0$$

$\therefore$ locus of P(h, k) is $y^2 - 4ax = (x + a)^2 \tan^2\theta_0$ **Ans.**

**Do yourself - 3 :**

**1.** Find the equation of the tangent to the parabola $y^2 = 12x$, which passes through the point (2, 5). Find also the co-ordinates of their points of contact.

**2.** Find the equation of the tangents to the parabola $y^2 = 16x$, which are parallel and perpendicular respectively to the line $2x - y + 5 = 0$. Find also the co-ordinates of their points of contact.

**3.** Prove that the locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ which meet at an angle $\theta$ is $(x + a)^2 \tan^2\theta = y^2 - 4ax$.

**4.** Write down the equations to the tangent at the point of the parabola $y^2 = 6x$ whose ordinate is 12.

**5.** Write down the equations to the tangent at the ends of the latus rectum of the parabola $y^2 = 12x$.

**6.** Find the equation to that tangent to the parabola $y^2 = 7x$ which is parallel to the straight line $4y - x + 3 = 0$. Find also its point of contact.

**7.** A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. Find its equation and its point of contact.

**8.** Find the equation to the tangents to the parabola $y^2 = 9x$ which goes through the point (4, 10).

**9.** Prove that the straight line $y = mx + c$ touches the parabola $y^2 = 4a(x + a)$ if $c = ma + \dfrac{a}{m}$

**10.** Prove that the straight line $\ell x + my + n = 0$ touches the parabola $y^2 = 4ax$ if $\ell n = am^2$.

**11.** Find the equations to the common tangents of the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is

**12.** Find the equations to the common tangents of the circle $x^2 + y^2 = 4ax$ and the parabola $y^2 = 4ax$ is

**13.** The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is

(A) $x^2 + 2y^2 - ax = 0$  (B) $2x^2 + y^2 - 2ax = 0$

(C) $2x^2 + 2y^2 - ay = 0$  (D) $2x^2 + y^2 - 2ay = 0$

**14.** Through the vertex O of the parabola, $y^2 = 4ax$ two chords OP & OQ are drawn and the circles on OP & OQ as diameters intersect in R. If $\theta_1, \theta_2$ & $\phi$ are the angles made with the axis by the tangents at P & Q on the parabola & by OR then the value of, $\cot\theta_1 + \cot\theta_2 =$

(A) $-2\tan\phi$ (B) $-2\tan(\pi - \phi)$ (C) 0 (D) $2\cot\phi$

**15.** Find the equations of the tangents to the parabola $y^2 = 16x$, which are parallel & perpendicular respectively to the line $2x - y + 5 = 0$. Find also the coordinates of their points of contact.

**16.** Find the equations of the tangents of the parabola $y^2 = 12x$, which passes through the point (2,5). Also find the point of contact.

**17.** Through the vertex O of the parabola $y^2 = 4ax$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q. Show that $OP \cdot OQ$ = constant.

**18.** Tangents are drawn to the parabola $y^2 = 12x$ at the points A,B and C such that the three tangents form a triangle PQR. If $\theta_1$, $\theta_2$ and $\theta_3$ be the inclinations of these tangents with the axis of x such that their cotangents form an arithmetical progression (in the same order) with common difference 2. Find the area of the triangle PQR.

**19.** Two tangents to the parabola $y^2 = 8x$ meet the tangent at its vertex in the points P & Q. If PQ = 4 units, prove that the locus of the point of the intersection of the two tangents is $y^2 = 8(x + 2)$.

**20.** Let a variable point A be lying on the directrix of parabola $y^2 = 4x$. Tangents AB & AC are drawn to the curve where B & C are points of contact of tangent. If the locus of centroid of $\Delta ABC$ is a conic then find the length of its latus rectum.

## 12. NORMAL TO THE PARABOLA $y^2 = 4ax$ :

**(a) Point form :**

Equation of normal to the given parabola at its point $(x_1, y_1)$ is

$$\mathbf{y - y_1 = -\frac{y_1}{2a}(x - x_1)}$$

**(b) Slope form :**

Equation of normal to the given parabola whose slope is 'm', is

$\mathbf{y = mx - 2am - am^3}$

foot of the normal is $\mathbf{(am^2, -2am)}$

**(c) Parametric form :**

Equation of normal to the given parabola at its point P(t), is

$\mathbf{y + tx = 2at + at^3}$

**Note :**

(i) Point of intersection of normals at $t_1$ & $t_2$ is $\mathbf{(a(t_1^2 + t_2^2 + t_1t_2 + 2), -at_1t_2(t_1 + t_2))}$.

(ii) If the normal to the parabola $y^2 = 4ax$ at the point $t_1$, meets the parabola again at the point $t_2$, then $\mathbf{t_2 = -\left(t_1 + \frac{2}{t_1}\right)}$.

(iii) If the normals to the parabola $y^2 = 4ax$ at the points $t_1$ & $t_2$ intersect again on the parabola at the point '$t_3$' then $\mathbf{t_1t_2 = 2\ ;\ t_3 = -(t_1 + t_2)}$ and the line joining $t_1$ & $t_2$ passes through a fixed point (–2a, 0).

(iv) If normal drawn to a parabola passes through a point P(h,k) then $k = mh - 2am - am^3$, i.e. $\mathbf{am^3 + m(2a - h) + k = 0}$.

This gives $\mathbf{m_1 + m_2 + m_3 = 0}$ ; $\mathbf{m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a}}$ ; $\mathbf{m_1m_2m_3 = \frac{-k}{a}}$

where $m_1$, $m_2$, & $m_3$ are the slopes of the three concurrent normals :

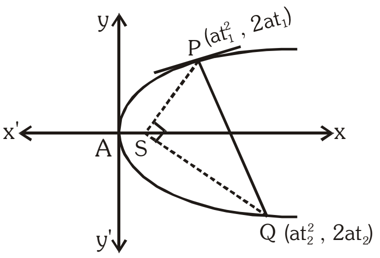
- Algebraic sum of slopes of the three concurrent normals is zero.
- Algebraic sum of ordinates of the three co-normal points on the parabola is zero.
- Centroid of the Δ formed by three co-normal points lies on the axis of parabola (x-axis).

***Illustration 11 :*** Prove that the normal chord to a parabola $y^2 = 4ax$ at the point whose ordinate is equal to abscissa subtends a right angle at the focus.

***Solution :*** Let the normal at $P(at_1^2, 2at_1)$ meet the curve at $Q(at_2^2, 2at_2)$

$\therefore$ PQ is a normal chord.

and $t_2 = -t_1 - \frac{2}{t_1}$ ..........(i)




By given condition $2at_1 = at_1^2$

$\therefore$ $t_1 = 2$ from equation (i), $t_2 = -3$

then P(4a, 4a) and Q(9a, –6a)

but focus S(a, 0)

$$\therefore \quad \text{Slope of SP} = \frac{4a-0}{4a-a} = \frac{4a}{3a} = \frac{4}{3}$$

$$\text{and} \quad \text{Slope of SQ} = \frac{-6a-0}{9a-a} = \frac{-6a}{8a} = -\frac{3}{4}$$

$$\therefore \quad \text{Slope of SP} \times \text{Slope of SQ} = \frac{4}{3} \times -\frac{3}{4} = -1$$

$\therefore$ $\angle PSQ = \pi/2$

i.e. PQ subtends a right angle at the focus S.

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***Illustration 12 :*** If two normals drawn from any point to the parabola $y^2 = 4ax$ make angle $\alpha$ and $\beta$ with the axis such that $\tan\alpha \, . \, \tan\beta = 2$, then find the locus of this point.

***Solution :*** Let the point is (h, k). The equation of any normal to the parabola $y^2 = 4ax$ is

$$y = mx - 2am - am^3$$

passes through (h, k)

$$k = mh - 2am - am^3$$

$$am^3 + m(2a - h) + k = 0 \qquad \text{....(i)}$$

$m_1, m_2, m_3$ are roots of the equation, then $m_1 . m_2 . m_3 = -\dfrac{k}{a}$

but $m_1m_2 = 2,\ m_3 = -\dfrac{k}{2a}$

$m_3$ is root of (i) $\quad \therefore \quad a\left(-\dfrac{k}{2a}\right)^3 - \dfrac{k}{2a}(2a-h) + k = 0 \ \Rightarrow\ k^2 = 4ah$

Thus locus is $y^2 = 4ax$. **Ans.**

***Illustration 13 :*** Three normals are drawn from the point (14, 7) to the curve $y^2 - 16x - 8y = 0$. Find the coordinates of the feet of the normals.

***Solution :*** The given parabola is $y^2 - 16x - 8y = 0$ ......... (i)

Let the co-ordinates of the feet of the normal from (14, 7) be $P(\alpha, \beta)$. Now the equation of the tangent at $P(\alpha, \beta)$ to parabola (i) is

$y\beta - 8(x + \alpha) - 4(y + \beta) = 0$

or $\quad (\beta - 4)y = 8x + 8a + 4\beta$ ......... (ii)

Its slope $= \dfrac{8}{\beta - 4}$

Equation of the normal to parabola (i) at $(\alpha, \beta)$ is $y - \beta = \dfrac{4-\beta}{8}(x - \alpha)$

It passes through (14, 7)

$\Rightarrow \quad 7 - \beta = \dfrac{4-\beta}{8}(14 - \alpha) \ \Rightarrow\ \alpha = \dfrac{6\beta}{\beta - 4}$ .......... (iii)

Also $(\alpha, \beta)$ lies on parabola (i) i.e. $\beta^2 - 16\alpha - 8\beta = 0$ ......... (iv)

Putting the value of $\alpha$ from (iii) in (iv), we get $\beta^2 - \dfrac{96\beta}{\beta - 4} - 8\beta = 0$

$\Rightarrow \quad \beta^2(\beta - 4) - 96\beta - 8\beta(\beta - 4) = 0 \Rightarrow \ \beta(\beta^2 - 4\beta - 96 - 8\beta + 32) = 0$

$\Rightarrow \quad \beta(\beta^2 - 12\beta - 64) = 0 \qquad \Rightarrow \quad \beta(\beta - 16)(\beta + 4) = 0$

$\Rightarrow \quad \beta = 0, 16, -4$

from (iii), $\alpha = 0$ when $\beta = 0$; $\alpha = 8$, when $\beta = 16$ ; $\alpha = 3$ when $\beta = -4$

Hence the feet of the normals are (0, 0), (8, 16) and (3, –4) **Ans.**

**Do yourself - 4 :**

1. If three distinct and real normals can be drawn to $y^2 = 8x$ from the point (a, 0), then -
(A) $a > 2$ (B) $a \in (2, 4)$ (C) $a > 4$ (D) none of these

2. Find the number of distinct normal that can be drawn from (–2, 1) to the parabola $y^2 - 4x - 2y - 3 = 0$.

3. If $2x + y + k = 0$ is a normal to the parabola $y^2 = -16x$, then find the value of k.

4. Three normals are drawn from the point (7, 14) to the parabola $x^2 - 8x - 16y = 0$. Find the co-ordinates of the feet of the normals.

5. Write down the equations to the normal at the point (4, 6) of the parabola $y^2 = 9x$

6. Write down the equations to the normal at the ends of the latus rectum of the parabola $y^2 = 4a\ (x - a)$

7. Find the points of the parabola $y^2 = 4ax$ at which the normal is inclined at 30° to the axis.

8. If two normals to a parabola $y^2 = 4ax$ intersect at right angles then the chord joining their feet pass through a fixed point whose co-ordinates are :
(A) (–2a,0) (B) (a,0) (C) (2a,0) (D) none

9. If the normal to a parabola $y^2 = 4ax$ at P meets the curve again in Q and if PQ and the normal at Q makes angles $\alpha$ and $\beta$ respectively with the x-axis then $\tan\alpha\,(\tan\alpha + \tan\beta)$ has the value equal to
(A) 0 (B) –2 (C) $-\frac{1}{2}$ (D) –1

10. If a normal to a parabola $y^2 = 4ax$ makes an angle $\phi$ with its axis, then it will cut the curve again at an angle
(A) $\tan^{-1}(2\tan\phi)$ (B) $\tan^{-1}\left(\frac{1}{2}\tan\phi\right)$ (C) $\cot^{-1}\left(\frac{1}{2}\tan\phi\right)$ (D) none

11. The tangent and normal at P(t), for all real positive t, to the parabola $y^2 = 4ax$ meet the axis of the parabola in T and G respectively, then the angle at which the tangent at P to the parabola is inclined to the tangent at P to the circle passing through the points P, T and G is
(A) $\cot^{-1}t$ (B) $\cot^{-1}t^2$ (C) $\tan^{-1}t$ (D) $\tan^{-1}t^2$

12. Normal to the parabola $y^2 = 8x$ at the point P (2, 4) meets the parabola again at the point Q. If C is the centre of the circle described on PQ as diameter then the coordinates of the image of the point C in the line $y = x$ are
(A) (– 4, 10) (B) (– 3, 8) (C) (4, – 10) (D) (– 3, 10)

13. PQ is a normal chord of the parabola $y^2 = 4ax$ at P, A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the x-axis in R. Then the length of AR is -
(A) equal to the length of the latus rectum.
(B) equal to the focal distance of the point P.
(C) equal to twice the focal distance of the point P.
(D) equal to the distance of the point P from the directrix.

**14.** Normals are drawn at points A, B, and C on the parabola $y^2 = 4x$ which intersect at P(h, k). The locus of the point P if the slope of the line joining the feet of two of them is 2 , is

(A) $x + y = 1$ (B) $x - y = 3$ (C) $y^2 = 2(x - 1)$ (D) $y^2 = 2\left(x - \frac{1}{2}\right)$

**15.** Three normals to $y^2 = 4x$ pass through the point (15, 12). Show that if one of the normals is given by $y = x - 3$ & find the equations of the others.

**16.** Prove that, the normal to $y^2 = 12x$ at (3, 6) meets the parabola again in (27, –18) & circle on this normal chord as diameter is $x^2 + y^2 - 30x + 12y - 27 = 0$.

**17.** From the point P(h, k) three normals are drawn to the parabola $x^2 = 8y$ and $m_1$, $m_2$ and $m_3$ are the slopes of three normals

(a) Find the algebraic sum of the slopes of these three normals.

(b) If two of the three normals are at right angles then the locus of point P is a conic, find the latus rectum of conic.

(c) If the two normals from P are such that they make complementary angles with the axis then the locus of point P is a conic, find a directrix of conic.

**18.** PC is the normal at P to the parabola $y^2 = 4ax$, C being on the axis. CP is produced outwards to Q so that PQ = CP; show that the locus of Q is a parabola.

**19.** Let P(a,b) and Q(c,d) are the two points on the parabola $y^2 = 8x$ such that the normals at them meet in (18,12). Find the product (abcd).

**20.** Find the condition on 'a' & 'b' so that the two tangents drawn to the parabola $y^2 = 4ax$ from a point are normals to the parabola $x^2 = 4by$.

## 13. AN IMPORTANT CONCEPT :

If a family of straight lines can be represented by an equation $\boldsymbol{\lambda^2 P + \lambda Q + R = 0}$ where $\lambda$ is a parameter and P, Q, R are linear functions of x and y then the family of lines will be tangent to the curve $\boldsymbol{Q^2 = 4PR}$.

***Illustration 14 :*** If the equation $m^2(x + 1) + m(y - 2) + 1 = 0$ represents a family of lines, where 'm' is parameter then find the equation of the curve to which these lines will always be tangents.

***Solution :*** $m^2(x + 1) + m(y - 2) + 1 = 0$

The equation of the curve to which above lines will always be tangents can be obtained by equating its discriminant to zero.

$\therefore \quad (y - 2)^2 - 4(x + 1) = 0$

$y^2 - 4y + 4 - 4x - 4 = 0$

$y^2 = 4(x + y)$ **Ans.**

**14. PAIR OF TANGENTS :**

The equation of the pair of tangents which can be drawn from any point $P(x_1, y_1)$ outside the parabola to the parabola $y^2 = 4ax$ is given by : $\mathbf{SS_1 = T^2}$ where :

$\mathbf{S \equiv y^2 - 4ax}$ ; $\mathbf{S_1 \equiv y_1^2 - 4ax_1}$ ; $\mathbf{T \equiv yy_1 - 2a\ (x + x_1)}$.

**15. DIRECTOR CIRCLE :**

Locus of the point of intersection of the perpendicular tangents to the parabola $y^2 = 4ax$ is called the **director circle.** It's equation is $\mathbf{x + a = 0}$ which is parabola's own directrix.

***Illustration 15 :*** The angle between the tangents drawn from a point $(-a, 2a)$ to $y^2 = 4ax$ is -

(A) $\pi/4$ (B) $\pi/2$ (C) $\pi/3$ (D) $\pi/6$

***Solution :*** The given point $(-a, 2a)$ lies on the directrix $x = -a$ of the parabola $y^2 = 4ax$. Thus, the tangents are at right angle. **Ans.(B)**

***Illustration 16 :*** The circle drawn with variable chord $x + ay - 5 = 0$ ($a$ being a parameter) of the parabola $y^2 = 20x$ as diameter will always touch the line -

(A) $x + 5 = 0$ (B) $y + 5 = 0$ (C) $x + y + 5 = 0$ (D) $x - y + 5 = 0$

***Solution :*** Clearly $x + ay - 5 = 0$ will always pass through the focus of $y^2 = 20x$ i.e. $(5, 0)$. Thus the drawn circle will always touch the directrix of the parabola i.e. the line $x + 5 = 0$. **Ans.(A)**

**Do yourself - 5 :**

**1.** If the equation $\lambda^2x + \lambda y - \lambda^2 + 2\lambda + 7 = 0$ represents a family of lines, where '$\lambda$' is parameter, then find the equation of the curve to which these lines will always be tangents.

**2.** Find the angle between the tangents drawn from the origin to the parabola, $y^2 = 4a(x - a)$.

**3.** Tangents are drawn from the point $(-1, 2)$ on the parabola $y^2 = 4x$. The length , these tangents will intercept on the line $x = 2$ is :

(A) 6 (B) $6\sqrt{2}$ (C) $2\sqrt{6}$ (D) none of these

**4.** Find locus of points of intersection of perpendicular tangents to parabolas.

(i) $(y - 4)^2 = 4(x - 3)$ (ii) $x^2 - 4x - 8y + 12 = 0$

**5.** Find combined equation of pair of tangents drawn to parabola.

(i) $y^2 = 8x$ from $(-3,1)$ (ii) $(x - 1)^2 = 4(y - 2)$ from $(2,0)$

**16. CHORD OF CONTACT :**

Equation of the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $\mathbf{yy_1 = 2a(x + x_1)}$

**Note :** The area of the triangle formed by the tangents from the point $(x_1, y_1)$ & the chord of contact is $\dfrac{(y_1^2 - 4ax_1)^{3/2}}{2a}$ i.e. $\dfrac{(S_1)^{3/2}}{2a}$, **also note that the chord of contact exists only if the point P is not inside.**

***Illustration 17 :*** If the line $x - y - 1 = 0$ intersect the parabola $y^2 = 8x$ at P & Q, then find the point of intersection of tangents at P & Q.

***Solution :*** Let (h, k) be point of intersection of tangents then chord of contact is

$yk = 4(x + h)$

$4x - yk + 4h = 0$ ....... (i)

But given line is

$x - y - 1 = 0$ ....... (ii)

Comparing (i) and (ii)

$$\therefore \quad \frac{4}{1} = \frac{-k}{-1} = \frac{4h}{-1} \quad \Rightarrow \quad h = -1, k = 4$$

$\therefore$ point $\equiv (-1, 4)$ **Ans.**

***Illustration 18 :*** Find the locus of point whose chord of contact w.r.t. to the parabola $y^2 = 4bx$ is the tangent of the parabola $y^2 = 4ax$.

***Solution :*** Equation of tangent to $y^2 = 4ax$ is $y = mx + \frac{a}{m}$ ......... (i)

Let it is chord of contact for parabola $y^2 = 4bx$ w.r.t. the point P(h, k)

$\therefore$ Equation of chord of contact is $yk = 2b(x + h)$

$$y = \frac{2b}{k}x + \frac{2bh}{k} \quad \text{......... (ii)}$$

From (i) & (ii)

$$m = \frac{2b}{k}, \frac{a}{m} = \frac{2bh}{k} \Rightarrow a = \frac{4b^2h}{k^2}$$

locus of P is $y^2 = \frac{4b^2}{a}x$. **Ans.**

## 17. CHORD WITH A GIVEN MIDDLE POINT :

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is $(x_1, y_1)$ is $\mathbf{y - y_1 = \frac{2a}{y_1}(x - x_1)}$.

This reduced to $\mathbf{T = S_1}$, where $\mathbf{T \equiv yy_1 - 2a\,(x + x_1)}$ & $\mathbf{S_1 \equiv y_1^2 - 4ax_1}$.

***Illustration 19 :*** Find the locus of middle point of the chord of the parabola $y^2 = 4ax$ which pass through a given (p, q).

***Solution :*** Let P(h, k) be the mid point of chord of the parabola $y^2 = 4ax$,

so equation of chord is $yk - 2a(x + h) = k^2 - 4ah$.

Since it passes through (p, q)

$\therefore$ $qk - 2a(p + h) = k^2 - 4ah$

$\therefore$ Required locus is $y^2 - 2ax - qy + 2ap = 0$.

***Illustration 20 :*** Find the locus of the middle point of a chord of a parabola $y^2 = 4ax$ which subtends a right angle at the vertex.

***Solution :*** The equation of the chord of the parabola whose middle point is $(\alpha, \beta)$ is

$y\beta - 2a(x + \alpha) = \beta^2 - 4a\alpha$

$\Rightarrow \quad y\beta - 2ax = \beta^2 - 2a\alpha$

or $\quad \dfrac{y\beta - 2ax}{\beta^2 - 2a\alpha} = 1 \qquad \qquad$ ......... (i)

Now, the equation of the pair of the lines OP and OQ joining the origin O i.e. the vertex to the points of intersection P and Q of the chord with the parabola $y^2 = 4ax$ is obtained by making the equation homogeneous by means of (i). Thus the equation of lines OP and OQ is $y^2 = \dfrac{4ax(y\beta - 2ax)}{\beta^2 - 2a\alpha}$

$\Rightarrow \quad y^2(\beta^2 - 2a\alpha) - 4a\beta xy + 8a^2x^2 = 0$

If the lines OP and OQ are at right angles, then the coefficient of $x^2$ + the coefficient of $y^2 = 0$

Therefore, $\beta^2 - 2a\alpha + 8a^2 = 0 \Rightarrow \beta^2 = 2a(\alpha - 4a)$

Hence the locus of $(\alpha, \beta)$ is $y^2 = 2a(x - 4a)$

**Do yourself - 6 :**

**1.** Find the equation of the chord of contacts of tangents drawn from a point (2, 1) to the parabola $x^2 = 2y$.

**2.** Find the co-ordinates of the middle point of the chord of the parabola $y^2 = 16x$, the equation of which is $2x - 3y + 8 = 0$

**3.** Find the locus of the mid-point of the chords of the parabola $y^2 = 4ax$ such that tangent at the extremities of the chords are perpendicular.

**4.** What is the equation to the chord of the parabola $y^2 = 8x$ which is bisected at the point (2, –3)?

**5.** Prove that the length of the chord joining the points of contact of tangents drawn from the point $(x_1,y_1)$ is $\dfrac{\sqrt{y_1^2 + 4a^2}\sqrt{y_1^2 - 4ax_1}}{a}$

**6.** TP & TQ are tangents to the parabola, $y^2 = 4ax$ at P & Q. If the chord PQ passes through the fixed point $(-a, b)$ then the locus of T is :

(A) $ay = 2b(x - b)$ (B) $bx = 2a(y - a)$

(C) $by = 2a(x - a)$ (D) $ax = 2b(y - b)$

7. Through the vertex O of a parabola $y^2 = 4x$, chords OP & OQ are drawn at right angles to one another. Show that for all positions of P, PQ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of PQ.

8. From the point $(-1, 2)$ tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.

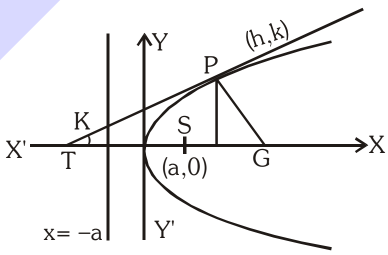
9. A variable chords of the parabola $y^2 = 8x$ touches the parabola $y^2 = 2x$. The locus of the point of intersection of the tangent at the end of the chord is a parabola. Find its latus rectum.

10. PQ, a variable chord of the parabola $y^2 = 4x$ subtends a right angle at the vertex. The tangents at P and Q meet at T and the normals at those points meet at N. If the locus of the mid point of TN is a parabola, then find its latus rectum.

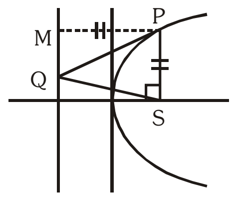
## 18. DIAMETER :

The locus of the middle points of a system of parallel chords of a Parabola is called a DIAMETER. Equation to the diameter of a parabola is **$y = 2a/m$**, where **$m$ = slope of parallel chords**.

## 19. IMPORTANT HIGHLIGHTS :




(a) If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then ST = SG = SP where 'S' is the focus. In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of the parabola after reflection.



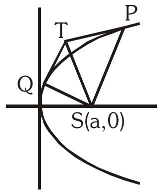

(b) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the **focus**.

(c) The tangents at the extremities of a focal chord intersect at right angles on the **directrix,** and a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P $(at^2, 2at)$ as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1+t^2}$ on a normal at the point P.

(d) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.

(e) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola is ; $\mathbf{2a} = \dfrac{2bc}{b+c}$ **i.e.** $\dfrac{1}{b}+\dfrac{1}{c}=\dfrac{1}{a}$.

**(f)** If the tangents at P and Q meet in T, then :

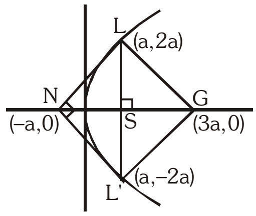



(i) TP and TQ subtend equal angles at the focus S.

(ii) $ST^2 = SP \cdot SQ$ &

(iii) The triangles SPT and STQ are similar.

**(g)** Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square, their points of intersection being (–a, 0) & (3a, 0).




**Note :**

(i) The two tangents at the extremities of focal chord meet on the foot of the directrix.

(ii) Figure LNL'G is square of side $2\sqrt{2}a$

**(h)** The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.

**Do yourself - 7 :**

**1.** The parabola $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the line x = 4, y = 4 and the co-ordinates axes. If $S_1$, $S_2$, $S_3$ are respectively the areas of these parts numbered from top to bottom; then find $S_1 : S_2 : S_3$.

**2.** Let P be the point (1, 0) and Q a point on the parabola $y^2 = 8x$, then find the locus of the mid point of PQ.

**3.** Prove that the normal chord at the point whose ordiante is equal to its abscissa subtends a right angle at the focus.

**4.** A chord is a normal to a parabola and is inclined at an angle θ to the axis; prove that the area of the triangle formed by it and the tangents at its extremities is $4a^2\sec^3\theta\ \text{cosec}^3\theta$.

**5.** Prove that the locus of the middle points of all tangents drawn from points on the directrix to the parabola is $y^2 (2x + a) = a\{3x + a)^2$.

**6.** A variable circle passes through the point A(2,1) and touches the x-axis. Locus of the other end of the diameter through A is a parabola.

(a) Find the length of the latus rectum of the parabola.

(b) Find the co-ordinates of the foot of the directrix of the parabola.

(c) The two tangents and two normals at the extremities of the latus rectum of the parabola constitutes a quadrilateral. Find area of quadrilateral.

**7.** Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is S(1, 2). If the length of latus-rectum of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (where m and n are coprime) then find the value of $(m + n)$.

## *Miscellaneous Illustrations :*

***Illustration 21 :*** The common tangent of the parabola $y^2 = 8ax$ and the circle $x^2 + y^2 = 2a^2$ is -

(A) $y = x + a$ (B) $x + y + a = 0$ (C) $x + y + 2a = 0$ (D) $y = x + 2a$

***Solution :*** Any tangent to parabola is $y = mx + \dfrac{2a}{m}$

Solving with the circle $x^2 + (mx + \dfrac{2a}{m})^2 = 2a^2 \Rightarrow x^2(1 + m^2) + 4ax + \dfrac{4a^2}{m^2} - 2a^2 = 0$

$B^2 - 4AC = 0$ gives $m = \pm 1$

Tangent $y = \pm x \pm 2a$ **Ans. (C,D)**

***Illustration 22 :*** If the tangent to the parabola $y^2 = 4ax$ meets the axis in T and tangent at the vertex A in Y and the rectangle TAYG is completed, show that the locus of G is $y^2 + ax = 0$.

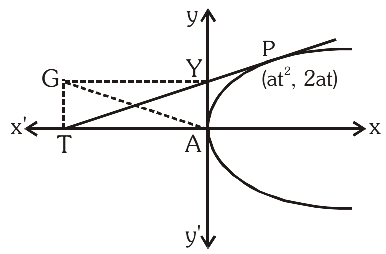
***Solution :*** Let $P(at^2, 2at)$ be any point on the parabola $y^2 = 4ax$.

Then tangent at $P(at^2, 2at)$ is $ty = x + at^2$

Since tangent meet the axis of parabola in T and tangent at the vertex in Y.

$\therefore$ Co-ordinates of T and Y are $(-at^2, 0)$ and $(0, at)$ respectively.

Let co-ordinates of G be $(x_1, y_1)$.




Since TAYG is rectangle.

$\therefore$ Mid-points of diagonals TY and GA is same

$$\Rightarrow \frac{x_1 + 0}{2} = \frac{-at^2 + 0}{2} \Rightarrow x_1 = -at^2 \qquad \text{.......... (i)}$$

and $$\frac{y_1 + 0}{2} = \frac{0 + at}{2} \Rightarrow y_1 = at \qquad \text{.......... (ii)}$$

Eliminating t from (i) and (ii) then we get $x_1 = -a\left(\dfrac{y_1}{a}\right)^2$

or $y_1^2 = -ax_1$ or $y_1^2 + ax_1 = 0$

$\therefore$ The locus of $G(x_1, y_1)$ is $y^2 + ax = 0$

***Illustration 23 :*** If P(–3, 2) is one end of the focal chord PQ of the parabola $y^2 + 4x + 4y = 0$, then the slope of the normal at Q is -

(A) –1/2 (B) 2 (C) 1/2 (D) –2

***Solution :*** The equation of the tangent at (–3, 2) to the parabola $y^2 + 4x + 4y = 0$ is

$2y + 2(x - 3) + 2(y + 2) = 0$

or $2x + 4y - 2 = 0 \Rightarrow x + 2y - 1 = 0$

Since the tangent at one end of the focal chord is parallel to the normal at the other end, the slope of the normal at the other end of the focal chord is $-\frac{1}{2}$. **Ans.(A)**

***Illustration 24 :*** Prove that the two parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ cannot have common normal, other than the axis unless $b/(a - c) > 2$.

***Solution :*** Given parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ have common normals. Then equation of normals in terms of slopes are $y = mx - 2am - am^3$ and $y = m(x - b) - 2cm - cm^3$ respectively then normals must be identical, compare the co-efficients

$$1 = \frac{2am + am^3}{mb + 2cm + cm^3}$$

$\Rightarrow \quad m[(c - a)m^2 + (b + 2c - 2a)] = 0,\ m \neq 0$ ($\because$ other than axis)

and $\quad m^2 = \dfrac{2a - 2c - b}{c - a},\ m = \pm\sqrt{\dfrac{2(a-c) - b}{c - a}}$

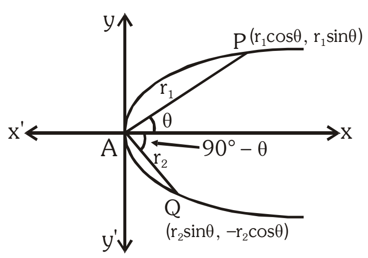
or $\quad m = \pm\sqrt{\left(-2 - \dfrac{b}{c - a}\right)}$

$\therefore \quad -2 - \dfrac{b}{c - a} > 0$

or $\quad -2 + \dfrac{b}{a - c} > 0 \Rightarrow \dfrac{b}{a - c} > 2$

***Illustration 25 :*** If $r_1$, $r_2$ be the length of the perpendicular chords of the parabola $y^2 = 4ax$ drawn through the vertex, then show that $(r_1 r_2)^{4/3} = 16a^2\left(r_1^{2/3} + r_2^{2/3}\right)$.

***Solution :*** Since chord are perpendicular, therefore if one makes an angle θ then the other will make an angle (90° – θ) with x-axis




Let $\quad AP = r_1$ and $AQ = r_2$

If $\quad \angle PAX = \theta$

then $\quad \angle QAX = 90° - \theta$

$\therefore \quad$ Co-ordinates of P and Q are $(r_1 \cos\theta,\ r_1 \sin\theta)$

and $\quad (r_2 \sin\theta,\ -r_2 \cos\theta)$ respectively.

Since P and Q lies on $y^2 = 4ax$

$$\therefore \quad r_1^2 \sin^2\theta = 4ar_1\cos\theta \text{ and } r_2^2\cos^2\theta = 4ar_2\sin\theta$$

$$\Rightarrow \quad r_1 = \frac{4a\cos\theta}{\sin^2\theta} \text{ and } r_2 = \frac{4a\sin\theta}{\cos^2\theta}$$

$$\therefore \quad (r_1r_2)^{4/3} = \left(\frac{4a\cos\theta}{\sin^2\theta}.\frac{4a\sin\theta}{\cos^2\theta}\right)^{4/3} = \left(\frac{16a^2}{\sin\theta\cos\theta}\right)^{4/3} \qquad \text{....... (i)}$$

$$\text{and} \quad 16a^2.\left(r_1^{2/3}+r_2^{2/3}\right) = 16a^2\left\{\left(\frac{4a\cos\theta}{\sin^2\theta}\right)^{2/3}+\left(\frac{4a\sin\theta}{\cos^2\theta}\right)^{2/3}\right\}$$

$$= 16a^2.(4a)^{2/3}\left\{\frac{(\cos\theta)^{2/3}}{(\sin\theta)^{4/3}}+\frac{(\sin\theta)^{2/3}}{(\cos\theta)^{4/3}}\right\} = 16a^2.(4a)^{2/3}\left\{\frac{\cos^2\theta+\sin^2\theta}{(\sin\theta)^{4/3}(\cos\theta)^{4/3}}\right\}$$

$$= \frac{16a^2.(4a)^{2/3}}{(\sin\theta\cos\theta)^{4/3}} = \left(\frac{16a^2}{\cos\theta\cos\theta}\right)^{4/3} = (r_1r_2)^{4/3} \qquad \{\text{from (i)}\}$$

***Illustration 26 :*** The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

***Solution :*** Let the three points on the parabola be

$(at_1^2, 2at_1), (at_2^2, 2at_2)$ and $(at_3^2, 2at_3)$

The area of the triangle formed by these points

$$\Delta_1 = \frac{1}{2}[at_1^2(2at_2 - 2at_3) + at_2^2(2at_3 - 2at_1) + at_2^2(2a_1 - 2at_2)]$$

$$= -a^2(t_2 - t_3)(t_3 - t_1)(t_1 - t_2).$$

The points of intersection of the tangents at these points are

$(at_2t_3, a(t_2+t_3)), (at_3t_1, a(t_3+t_1))$ and $(at_1t_2, a(t_1+t_2))$

The area of the triangle formed by these three points

$$\Delta_2 = \frac{1}{2}\{at_2t_3(at_3 - at_2) + at_3t_1(at_1 - at_3) + at_1t_2(at_2 - at_1)\}$$

$$= \frac{1}{2}a^2(t_2 - t_3)(t_3 - t_1)(t_1 - t_2)$$

Hence $\Delta_1 = 2\Delta_2$

***Illustration 27 :*** Prove that the orthocentre of any triangle formed by three tangents to a parabola lies on the directrix.

***Solution :*** Let the equations of the three tangents be

$t_1y = x + at_1^2$ ................(i)

$t_2y = x + at_2^2$ ................(ii)

and $t_3y = x + at_3^2$ ................(iii)

The point of intersection of (ii) and (iii) is found, by solving them, to be $(at_2t_3, a(t_2 + t_3))$

The equation of the straight line through this point & perpendicular to (i) is

$y - a(t_2 + t_3) = -t_1(x - at_2t_3)$

i.e. $y + t_1x = a(t_2 + t_3 + t_1t_2t_3)$ ................(iv)

Similarly, the equation of the straight line through the point of intersection of (iii) and (i) & perpendicular to (ii) is

$y + t_2x = a(t_3 + t_1 + t_1t_2t_3)$ ................(v)

and the equation of the straight line through the point of intersection of (i) and (ii) & perpendicular to (iii) is

$y + t_1x = a(t_1 + t_2 + t_1t_2t_3)$ ................(vi)

The point which is common to the straight lines (iv), (v) and (vi)

i.e. the orthocentre of the triangle, is easily seen to be the point whose coordinates are

$x = -a,\ y = a(t_1 + t_2 + t_3 + t_1t_2t_3)$

and this point lies on the directrix.

# ELLIPSE

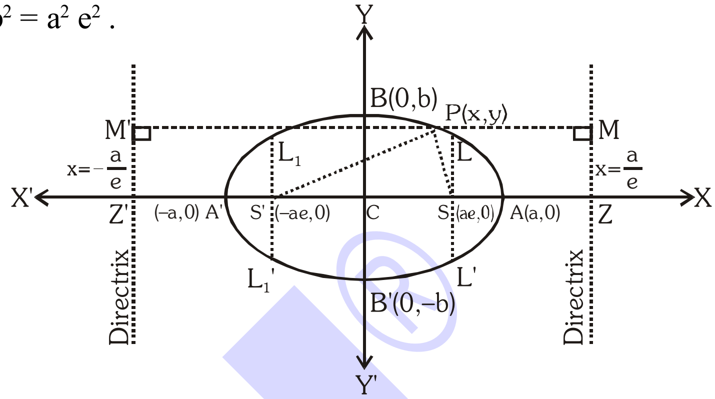
## 1. STANDARD EQUATION & DEFINITION :

Standard equation of an ellipse referred to its principal axes along the co-ordinate axes is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

where **a > b & $b^2 = a^2(1-e^2)$** $\Rightarrow a^2 - b^2 = a^2e^2$.

where e = eccentricity **(0 < e < 1)**.

FOCI : **S ≡ (ae, 0) & S' ≡ (– ae, 0 )**.




**(a) Equation of directrices :**

$$\mathbf{x} = \frac{a}{e} \quad \& \quad x = -\frac{a}{e}.$$

**(b) Vertices :**

**A' ≡ (–a, 0) & A ≡ (a, 0).**

**(c) Major axis :** The line segment A' A in which the foci S' & S lie is of length 2a & is called the **major axis** (a > b) of the ellipse. Point of intersection of major axis with directrix is called **the foot of the directrix (z)** $\left(\pm\frac{a}{e},\ 0\right)$.

**(d) Minor Axis :** The y-axis intersects the ellipse in the points B' ≡ (0,- b) & B ≡ (0, b). The line segment B'B of length 2b (b < a) is called the **Minor Axis** of the ellipse.

**(e) Principal Axes :** The major & minor axis together are called **Principal Axes** of the ellipse.

**(f) Centre :** The point which bisects every chord of the conic drawn through it is called the **centre** of the conic. C ≡ (0,0) the origin is the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

**(g) Diameter :** A chord of the conic which passes through the centre is called a **diameter** of the conic.

**(h) Focal Chord :** A chord which passes through a focus is called a **focal chord.**

**(i) Double Ordinate :** A chord perpendicular to the major axis is called a **double ordinate.**

**(j) Latus Rectum :** The focal chord perpendicular to the major axis is called the **latus rectum**.

(i) Length of latus rectum (LL') $=\frac{2b^2}{a}=\frac{(\text{minor axis})^2}{\text{major axis}}=2a(1-e^2)$

(ii) Equation of latus rectum : **x = ± ae**.

(iii) Ends of the latus rectum are $L\left(ae,\frac{b^2}{a}\right), L'\left(ae,-\frac{b^2}{a}\right), L_1\left(-ae,\frac{b^2}{a}\right)$ **and** $L_1'\left(-ae,-\frac{b^2}{a}\right)$.

(k) **Focal radii : SP = a –ex & S'P = a + ex** $\Rightarrow$ **SP + S 'P = 2a = Major axis.**

(l) **Eccentricity :** $e=\sqrt{1-\frac{b^2}{a^2}}$

**Note :**

(i) The sum of the focal distances of any point on the ellipse is equal to the major Axis. Hence distance of focus from the extremity of a minor axis is equal to semi major axis. **i.e BS = CA.**

(ii) If the equation of the ellipse is given as $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ & nothing is mentioned, then the rule is to assume that a > b.

***Illustration 1 :*** If LR of an ellipse is half of its minor axis, then its eccentricity is -

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{2}}{3}$

***Solution :*** As given $\frac{2b^2}{a}=b$ $\Rightarrow$ $2b = a$ $\Rightarrow$ $4b^2 = a^2$

$\Rightarrow$ $4a^2(1-e^2) = a^2$ $\Rightarrow$ $1 - e^2 = 1/4$

$\therefore$ $e=\sqrt{3}/2$ **Ans. (C)**

***Illustration 2 :*** Find the equation of the ellipse whose foci are (2, 3), (–2, 3) and whose semi minor axis is of length $\sqrt{5}$.

***Solution :*** Here S is (2, 3) & S' is (–2, 3) and $b=\sqrt{5}$ $\Rightarrow$ SS' = 4 = 2ae $\Rightarrow$ ae = 2

but $b^2 = a^2(1-e^2)$ $\Rightarrow$ $5 = a^2 - 4$ $\Rightarrow$ a = 3.

Hence the equation to major axis is y = 3

Centre of ellipse is midpoint of SS' i.e. (0, 3)

$\therefore$ Equation to ellipse is $\frac{x^2}{a^2}+\frac{(y-3)^2}{b^2}=1$ or $\frac{x^2}{9}+\frac{(y-3)^2}{5}=1$ **Ans.**

***Illustration 3 :*** Find the equation of the ellipse having centre at (1, 2), one focus at (6, 2) and passing through the point (4, 6).

***Solution :*** With centre at (1, 2), the equation of the ellipse is $\frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1$. It passes through the point (4, 6)

$\Rightarrow$ $\frac{9}{a^2}+\frac{16}{b^2}=1$ .......... (i)

Distance between the focus and the centre = (6 – 1) = 5 = ae

$\Rightarrow$ $b^2 = a^2 - a^2e^2 = a^2 - 25$ .......... (ii)

Solving for $a^2$ and $b^2$ from the equations (i) and (ii), we get $a^2 = 45$ and $b^2 = 20$.

Hence the equation of the ellipse is $\frac{(x-1)^2}{45}+\frac{(y-2)^2}{20}=1$ **Ans.**

**Do yourself - 1 :**

**1.** If LR of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $(a<b)$ is half of its major axis, then find its eccentricity.

**2.** Find the equation of the ellipse whose foci are (4, 6) & (16, 6) and whose semi-minor axis is 4.

**3.** Find the eccentricity, foci and the length of the latus-rectum of the ellipse $x^2+4y^2+8y-2x+1=0$.

**4.** Find the equation to the ellipse, whose centres are the origin, whose axes are the axes of coordinates, and which pass through

(a) the points (2,2), and (3,1) (b) the points (1,4) and (–6,1)

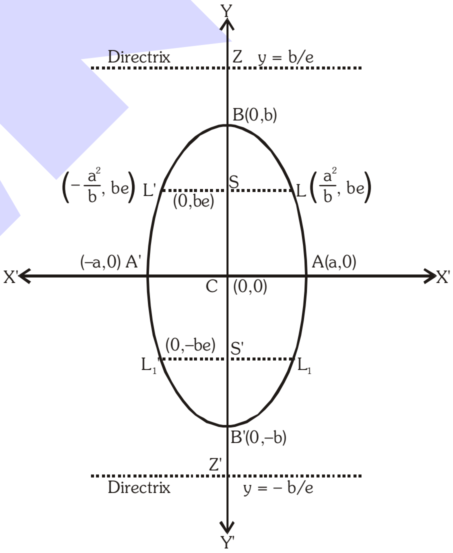
**5.** Find the equation of the ellipse with centre origin and axes as the axes of coordinates

(a) whose latus rectum is 5 and whose eccentricity is $\frac{2}{3}$,

(b) whose minor axis is equal to the distance between the foci and whose latus rectum is 10,

(c) Whose foci are the points (4,0) and (–4,0) and whose eccentricity is $\frac{1}{3}$

**2. ANOTHER FORM OF ELLIPSE :** $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, **(a <b)**




**(a)** **AA' = Minor axis = 2a**

**(b)** **BB' = Major axis = 2b**

**(c)** $a^2 = b^2(1-e^2)$

**(d)** **Latus rectum LL' = $L_1L_1'$** $=\frac{2a^2}{b}$, **equation y = ± be**

**(e)** **Ends of the latus rectum are :**

$$L\left(\frac{a^2}{b},be\right),L'\left(-\frac{a^2}{b},be\right),L_1\left(\frac{a^2}{b},-be\right),L_1'\left(-\frac{a^2}{b},-be\right)$$

**(f)** **Equation of directrix y = ± b/e**

**(g)** **Eccentricity** : $e=\sqrt{1-\frac{a^2}{b^2}}$

***Illustration 4 :*** The equation of the ellipse with respect to coordinate axes whose minor axis is equal to the distance between its foci and whose LR = 10, will be-

(A) $2x^2+y^2=100$ (B) $x^2+2y^2=100$ (C) $2x^2+3y^2=80$ (D) none of these

***Solution :*** When $a>b$

As given $2b=2ae \Rightarrow b=ae$ ...... (i)

Also $\frac{2b^2}{a}=10 \Rightarrow b^2=5a$ ...... (ii)

Now since $b^2=a^2-a^2e^2 \Rightarrow b^2=a^2-b^2$ [From (i)]

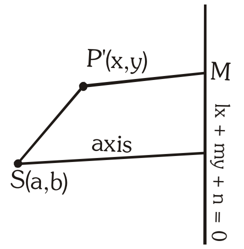
$\Rightarrow 2b^2=a^2$ ...... (iii)

(ii), (iii) $\Rightarrow a^2=100, b^2=50$

Hence equation of the ellipse will be $\frac{x^2}{100}+\frac{y^2}{50}=1 \Rightarrow x^2+2y^2=100$

Similarly when $a<b$ then required ellipse is $2x^2+y^2=100$ **Ans. (A, B)**

## 3. GENERAL EQUATION OF AN ELLIPSE




Let (a, b) be the focus S, and $lx + my + n = 0$ is the equation of directrix.

Let P(x, y) be any point on the ellipse. Then by definition.

$\Rightarrow$ **SP = e PM** (e is the eccentricity) $\Rightarrow (x-a)^2 + (y-b)^2 = e^2\dfrac{(lx+my+n)^2}{(l^2+m^2)}$

$\Rightarrow (l^2 + m^2)\{(x-a)^2 + (y-b)^2\} = e^2\{lx + my + n\}^2$

**Do yourself - 2 :**

**1.** The foci of an ellipse are $(0, \pm 2)$ and its eccentricity is $\dfrac{1}{\sqrt{2}}$. Find its equation

**2.** Find the centre, the length of the axes, eccentricity and the foci of ellipse
$12x^2 + 4y^2 + 24x - 16y + 25 = 0$

**3.** The equation $\dfrac{x^2}{8-t} + \dfrac{y^2}{t-4} = 1$, will represent an ellipse if

(A) $t \in (1, 5)$ (B) $t \in (2, 8)$ (C) $t \in (4, 8) - \{6\}$ (D) $t \in (4, 10) - \{6\}$

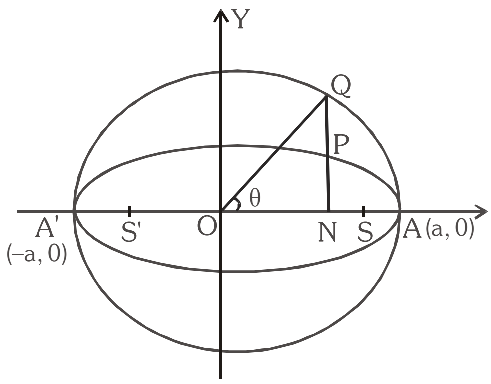
**4.** Find the equation to the ellipse, whose focus is the point (–1,1), whose directrix is the straight line $x - y + 3 = 0$, and whose eccentricity is $\dfrac{1}{2}$.

## 4. POSITION OF A POINT W.R.T. AN ELLIPSE :

The point **P($x_1$, $y_1$)** lies **outside**, **inside** or **on** the ellipse according as ; $\dfrac{x_1^2}{a^2} + \dfrac{y_1^2}{b^2} - 1$ **> < or = 0**.

## 5. AUXILLIARY CIRCLE/ECCENTRIC ANGLE :

A circle described on major axis as diameter is called the **auxiliary circle.** Let Q be a point on the auxiliary circle $x^2 + y^2 = a^2$ such that QP produced is perpendicular to the x-axis then P & Q are called as the CORRESPONDING POINTS on the ellipse & the auxiliary circle respectively.'θ' is called the **ECCENTRIC ANGLE** of the point P on the ellipse $(0 \le \theta < 2\pi)$.




Note that $\dfrac{l(PN)}{l(QN)} = \dfrac{b}{a} = \dfrac{\text{Semi minor axis}}{\text{Semi major axis}}$

Hence "If from each point of a circle perpendiculars are drawn upon a fixed diameter then the locus of the points dividing these perpendiculars in a given ratio is an ellipse of which the given circle is the auxiliary circle".

## 6. PARAMETRIC REPRESENTATION :

The equations $x = a\cos\theta$ & $y = b\sin\theta$ together represent the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

where $\theta$ is a parameter (eccentric angle).

Note that if $P(\theta) \equiv (a\cos\theta, b\sin\theta)$ is on the ellipse then ; $\mathbf{Q(\theta) \equiv (a\cos\theta, a\sin\theta)}$ is on the auxiliary circle.

## 7. LINE AND AN ELLIPSE :

The line $y = mx + c$ meets the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in two points real, coincident or imaginary according as $\mathbf{c^2}$ **is < = or >** $\mathbf{a^2m^2 + b^2}$.

Hence $y = mx + c$ is tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if $\mathbf{c^2 = a^2m^2 + b^2}$.

The equation to the chord of the ellipse joining two points with eccentric angles $\alpha$ & $\beta$ is given by $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$.

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***Illustration 5 :*** For what value of $\lambda$ does the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$.

***Solution :*** $\because$ Equation of ellipse is $9x^2 + 16y^2 = 144$ or $\frac{x^2}{16}+\frac{y^2}{9}=1$

Comparing this with $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ then we get $a^2 = 16$ and $b^2 = 9$

and comparing the line $y = x + \lambda$ with $y = mx + c$ $\quad\therefore\quad m = 1$ and $c = \lambda$

If the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$, then $c^2 = a^2m^2 + b^2$

$\Rightarrow \lambda^2 = 16\times 1^2 + 9 \quad \Rightarrow \lambda^2 = 25 \quad \therefore \lambda = \pm 5$ **Ans.**

***Illustration 6 :*** If $\alpha$, $\beta$ are eccentric angles of end points of a focal chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\tan\alpha/2.\ \tan\beta/2$ is equal to -

(A) $\frac{e-1}{e+1}$ (B) $\frac{1-e}{1+e}$ (C) $\frac{e+1}{e-1}$ (D) $\frac{1+e}{1-e}$

***Solution :*** Equation of line joining points '$\alpha$' and '$\beta$' is $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$

If it is a focal chord, then it passes through focus (ae, 0), so $e\cos\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$

$$\Rightarrow \frac{\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}}=\frac{e}{1} \quad \Rightarrow \frac{\cos\frac{\alpha-\beta}{2}-\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}+\cos\frac{\alpha+\beta}{2}}=\frac{e-1}{e+1}$$

$$\Rightarrow \frac{2\sin\alpha/2\ \sin\beta/2}{2\cos\alpha/2\ \cos\beta/2}=\frac{e-1}{e+1} \quad \Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{e-1}{e+1}$$

using (–ae, 0) , we get $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{e+1}{e-1}$ **Ans. (A,C)**

**Do yourself - 3 :**

**1.** Find the position of the point (4, 3) relative to the ellipse $2x^2 + 9y^2 = 113$.

**2.** A tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, (a > b) having slope –1 intersects the axis of x & y in point A & B respectively. If O is the origin then find the area of triangle OAB.

**3.** Find the condition for the line $x\cos\theta + y\sin\theta = P$ to be a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

**4.** Is the point (4,–3) within or without the ellipse $5x^2 + 7y^2 = 11^2$ ?

**5.** Find the lengths of and the equations to, the focal radii drawn to the point $\left(4\sqrt{3},5\right)$ of the ellipse $25x^2 + 16y^2 = 1600$.

**6.** Find the locus of the middle points of chords of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ which are drawn through the positive end of the minor axis.

**7.** Prove that the area of the triangle formed by three points on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose eccentric angles are $\theta,\phi$, and $\psi$, is $\left|2ab\sin\frac{\phi-\psi}{2}\sin\frac{\psi-\theta}{2}\sin\frac{\theta-\phi}{2}\right|$.

## 8. TANGENT TO THE ELLIPSE $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ :

**(a) Point form :** Equation of tangent to the given ellipse at its point $(x_1, y_1)$ is $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$

**Note :** For general ellipse replace $x^2$ by $(xx_1)$, $y^2$ by $(yy_1)$, $2x$ by $(x + x_1)$, $2y$ by $(y + y_1)$, $2xy$ by $(xy_1 + yx_1)$ & c by (c).

**(b) Slope form :** Equation of tangent to the given ellipse whose slope is 'm', is $\mathbf{y = mx \pm \sqrt{a^2m^2+b^2}}$

Point of contact are $\left(\frac{\mp a^2m}{\sqrt{a^2m^2+b^2}},\frac{\pm b^2}{\sqrt{a^2m^2+b^2}}\right)$

Note that there are two tangents to the ellipse having the same m, i.e. there are two tangents parallel to any given direction.

**(c) Parametric form :** Equation of tangent to the given ellipse at its point $(a\cos\theta, b\sin\theta)$, is

$$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$$

**Note :**

(i) The eccentric angles of point of contact of two parallel tangents differ by $\pi$.

(ii) Point of intersection of the tangents at the point $\alpha$ & $\beta$ is $\left(a\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, b\frac{\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$

***Illustration 7 :*** Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.

***Solution :*** Let m be the slope of the tangent, since the tangent is perpendicular to the line $y + 2x = 4$.

$\therefore \quad mx - 2 = -1 \Rightarrow \quad m = \frac{1}{2}$

Since $3x^2 + 4y^2 = 12$ or $\frac{x^2}{4} + \frac{y^2}{3} = 1$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\therefore \quad a^2 = 4$ and $b^2 = 3$

So the equation of the tangent are $y = \frac{1}{2}x \pm \sqrt{4 \times \frac{1}{4} + 3}$

$\Rightarrow \quad y = \frac{1}{2}x \pm 2$ or $\quad x - 2y \pm 4 = 0$. **Ans.**

***Illustration 8 :*** The tangent at a point P on an ellipse intersects the major axis in T and N is the foot of the perpendicular from P to the same axis. Show that the circle drawn on NT as diameter intersects the auxiliary circle orthogonally.

***Solution :*** Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let P(acosθ, bsinθ) be a point on the ellipse.

The equation of the tangent at P is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$. It meets the major axis at $T \equiv (a\sec\theta, 0)$.

The coordinates of N are $(a\cos\theta, 0)$. The equation of the circle with NT as its diameter is $(x - a\sec\theta)(x - a\cos\theta) + y^2 = 0$.

$\Rightarrow \quad x^2 + y^2 - ax(\sec\theta + \cos\theta) + a^2 = 0$

It cuts the auxiliary circle $x^2 + y^2 - a^2 = 0$ orthogonally if

$2g \,.\, 0 + 2f \,.\, 0 = a^2 - a^2 = 0$, which is true. **Ans.**

**Do yourself - 4 :**

**1.** Find the equation of the tangents to the ellipse $9x^2 + 16y^2 = 144$ which are parallel to the line $x + 3y + k = 0$.

**2.** Find the equation of the tangent to the ellipse $7x^2 + 8y^2 = 100$ at the point $(2, -3)$.

**3.** Find the equation to the tangent

(a) at the point $\left(1, \frac{4}{3}\right)$ of the ellipse $4x^2 + 9y^2 = 20$.

(b) at the point of the ellipse $5x^2 + 3y^2 = 137$ whose ordinate is 2

(c) at the ends of the latera recta of the ellipse $9x^2 + 16y^2 = 144$.

**4.** Prove that the straight line $y = x + \sqrt{\frac{7}{12}}$ touches the ellipse $3x^2 + 4y^2 = 1$.

**5.** Find the equations to the tangents to the ellipse $4x^2 + 3y^2 = 5$ which are parallel to the straight line $y = 3x + 7$. Also find points of tangency.

**6.** Find the equations to the tangents at the ends of the latera recta of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, and show that they pass through the intersections of the axis and the directrices.

**7.** Find the points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ such that the tangents there are equally inclined to the axes. Prove also that the length of the perpendicular from the centre on either of these tangents is $\sqrt{\frac{a^2+b^2}{2}}$

**8.** Prove that the locus of the middle points of the portions of tangents of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ included between the axes is the curve $\frac{a^2}{x^2}+\frac{b^2}{y^2}=4$.

## 9. NORMAL TO THE ELLIPSE $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ :

**(a)** **Point form :** Equation of the normal to the given ellipse at $(x_1, y_1)$ is

$$\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=\mathbf{a^2-b^2=a^2e^2}.$$

**(b)** **Slope form :** Equation of a normal to the given ellipse whose slope is 'm' is

$$\mathbf{y = mx} \mp \frac{(a^2-b^2)m}{\sqrt{a^2+b^2m^2}}.$$

**(c)** **Parametric form :** Equation of the normal to the given ellipse at the point **($a\cos\theta$, $b\sin\theta$)** is

$$\mathbf{ax\sec\theta - by\,\mathrm{cosec}\,\theta = (a^2-b^2)}.$$

***Illustration 9 :*** Find the condition that the line $\ell x + my = n$ may be a normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

***Solution :*** Equation of normal to the given ellipse at $(a\cos\theta, b\sin\theta)$ is $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$ ...(i)

If the line $\ell x + my = n$ is also normal to the ellipse then there must be a value of $\theta$ for which line (i) and line $\ell x + my = n$ are identical. For that value of $\theta$ we have

$$\frac{\ell}{\left(\frac{a}{\cos\theta}\right)}=\frac{m}{-\left(\frac{b}{\sin\theta}\right)}=\frac{n}{(a^2-b^2)} \quad \text{or} \quad \cos\theta=\frac{an}{\ell(a^2-b^2)} \qquad \text{...... (iii)}$$

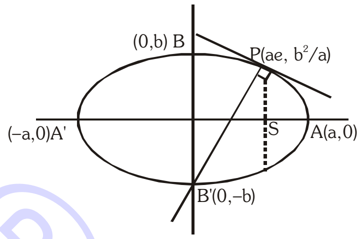
$$\text{and} \quad \sin\theta=\frac{-bn}{m(a^2-b^2)} \qquad \text{...... (iv)}$$

Squaring and adding (iii) and (iv), we get $1=\frac{n^2}{(a^2-b^2)^2}\left(\frac{a^2}{\ell^2}+\frac{b^2}{m^2}\right)$ which is the required condition.

***Illustration 10 :*** If the normal at an end of a latus-rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through one extremity of the minor axis, show that the eccentricity of the ellipse is given by $e=\sqrt{\frac{\sqrt{5}-1}{2}}$

***Solution :*** The co-ordinates of an end of the latus-rectum are $(ae, b^2/a)$.

The equation of normal at $P(ae, b^2/a)$ is




$$\frac{a^2x}{ae}-\frac{b^2(y)}{b^2/a}=a^2-b^2 \quad \text{or} \quad \frac{ax}{e}-ay=a^2-b^2$$

It passes through one extremity of the minor axis whose co-ordinates are $(0, -b)$

$\therefore \quad 0+ab=a^2-b^2 \quad \Rightarrow \quad (a^2b^2)=(a^2-b^2)^2$

$\Rightarrow \quad a^2.a^2(1-e^2)=(a^2e^2)^2 \quad \Rightarrow \quad 1-e^2=e^4$

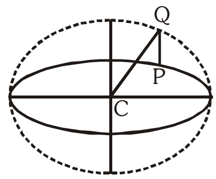
$\Rightarrow \quad e^4+e^2-1=0 \quad \Rightarrow \quad (e^2)^2+e^2-1=0$

$\therefore \quad e^2=\frac{-1\pm\sqrt{1+4}}{2} \quad \Rightarrow \quad e=\sqrt{\frac{\sqrt{5}-1}{2}}$ (taking positive sign) **Ans.**

***Illustration 11 :*** P and Q are corresponding points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the auxiliary circles respectively. The normal at P to the ellipse meets CQ in R, where C is the centre of the ellipse. Prove that CR = a + b

***Solution :*** Let $P \equiv (a\cos\theta, b\sin\theta)$

$\therefore \quad Q \equiv (a\cos\theta, a\sin\theta)$




Equation of normal at P is

$(a\sec\theta)x-(b\,\text{cosec}\theta)y=a^2-b^2$ ........... (i)

equation of CQ is $y=\tan\theta \,.\, x$ ........... (ii)

Solving equation (i) & (ii), we get $(a-b)x=(a^2-b^2)\cos\theta$

$x=(a+b)\cos\theta$, & $y=(a+b)\sin\theta$

$\therefore \quad R \equiv ((a+b)\cos\theta, (a+b)\sin\theta$

$\therefore \quad CR = a + b$ **Ans.**

**Do yourself - 5 :**

**1.** Find the equation of the normal to the ellipse $9x^2+16y^2=288$ at the point (4, 3)

**2.** Let P be a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with foci $F_1$ and $F_2$. If A is the area of the triangle $PF_1F_2$, then find maximum value of A.

**3.** If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{3}+\frac{y^2}{2}=1$ intersects it again at the point $Q(2\theta)$, then find $\cos\theta$.

**4.** Show that for all real values of 't' the line $2tx + y\sqrt{1-t^2} = 1$ touches a fixed ellipse. Find the eccentricity of the ellipse.

**5.** Find the equation to normal

(a) at the point $\left(1, \frac{4}{3}\right)$ of the ellipse $4x^2 + 9y^2 = 20$.

(b) at the point of the ellipse $5x^2 + 3y^2 = 137$ whose ordinate is 2

(c) at the ends of the latera recta of the ellipse $9x^2 + 16y^2 = 144$.

**6.** Find the equations to the normals of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the ends of the latera recta, and prove that each passes through an end of the minor axis if $e^4 + e^2 = 1$.

**7.** Prove that the straight line $\ell x + my = n$ is a normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, if $\frac{a^2}{\ell^2} + \frac{b^2}{m^2} = \frac{(a^2 - b^2)^2}{n^2}$

## 10. CHORD OF CONTACT :

If PA and PB be the tangents from point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The equation of the chord of contact AB is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ **or T = 0 (at $x_1, y_1$).**

***Illustration 12 :*** If tangents to the parabola $y^2 = 4ax$ intersect the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at A and B, the find the locus of point of intersection of tangents at A and B.

***Solution :*** Let $P \equiv (h, k)$ be the point of intersection of tangents at A & B

$\therefore$ equation of chord of contact AB is $\frac{xh}{a^2} + \frac{yk}{b^2} = 1$ .......... (i)

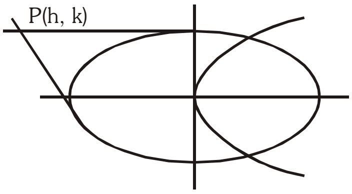
which touches the parabola.

Equation of tangent to parabola $y^2 = 4ax$ is $y = mx + \frac{a}{m}$

$\Rightarrow$ $mx - y = -\frac{a}{m}$ .......... (ii)

equation (i) & (ii) as must be same

$\therefore$ $\frac{m}{\left(\frac{h}{a^2}\right)} = \frac{-1}{\left(\frac{k}{b^2}\right)} = \frac{-\frac{a}{m}}{1} \Rightarrow m = -\frac{h}{k}\frac{b^2}{a^2}$ & $m = \frac{ak}{b^2}$




$\therefore$ $-\frac{hb^2}{ka^2} = \frac{ak}{b^2} \Rightarrow$ locus of P is $y^2 = -\frac{b^4}{a^3}.x$ **Ans.**

**Do yourself - 6 :**

**1.** Find the equation of chord of contact to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at the point (1, 3).

**2.** If the chord of contact of tangents from two points $(x_1, y_1)$ and $(x_2, y_2)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are at right angles, then find $\frac{x_1x_2}{y_1y_2}$.

**3.** If a line $3x - y = 2$ intersects ellipse $\frac{x^2}{8}+\frac{y^2}{4}=1$ at points A & B, then find co-ordinates of point of intersection of tangents at points A & B.

**4.** Tangents are drawn from the point (–2, 2) to the ellipse $x^2 + 2y^2 = 6$. Find the equation of chord of contact.

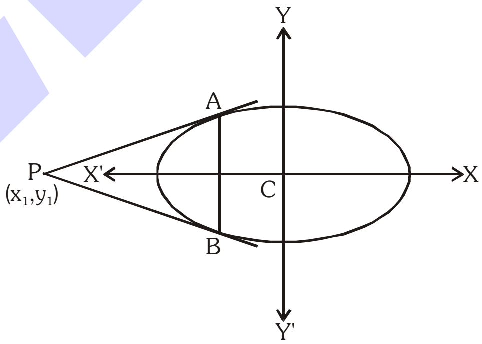
**5.** Line $3x + 4y = 12$ intersect the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ at P and Q. If the point of intersection of tangents at P and Q is $\left(\frac{25}{a},\frac{16}{b}\right)$, then a + b is equal to

## 11. PAIR OF TANGENTS :

If $P(x_1,y_1)$ be any point lies outside the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$,

and a pair of tangents PA, PB can be drawn to it from P.

Then the equation of pair of tangents of PA and PB is $SS_1 = T^2$




where $S_1=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1,\ T=\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1$

i.e. $\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)\left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1\right)=\left(\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1\right)^2$

## 12. DIRECTOR CIRCLE :

Locus of the point of intersection of the tangents which meet at right angles is called the **Director Circle.** The equation to this locus is $\mathbf{x^2 + y^2 = a^2 + b^2}$ i.e. a circle whose centre is the centre of the ellipse & whose radius is the length of the line joining the ends of the major & minor axis.

***Illustration 13 :*** A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P and Q. Prove that the tangents at P and Q of the ellipse $x^2 + 2y^2 = 6$ are at right angles.

***Solution :*** Given ellipse are $\frac{x^2}{4}+\frac{y^2}{1}=1$ ......... (i)

and, $\frac{x^2}{6}+\frac{y^2}{3}=1$ .......... (ii)

any tangent to (i) is $\frac{x\cos\theta}{2}+\frac{y\sin\theta}{1}=1$ ......... (iii)

It cuts (ii) at P and Q, and suppose tangent at P and Q meet at (h, k) Then equation of chord of contact of (h, k) with respect to ellipse (ii) is $\frac{hx}{6}+\frac{ky}{3}=1$ ......... (iv)

comparing (iii) and (iv), we get $\frac{\cos\theta}{h/3}=\frac{\sin\theta}{k/3}=1$

$\Rightarrow \cos\theta=\frac{h}{3}$ and $\sin\theta=\frac{k}{3} \Rightarrow h^2+k^2=9$

locus of the point (h, k) is $x^2+y^2=9 \Rightarrow x^2+y^2=6+3=a^2+b^2$

i.e. director circle of second ellipse. Hence the tangents are at right angles.

## 13. EQUATION OF CHORD WITH MID POINT $(x_1,y_1)$ :

The equation of the chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose mid-point be $(x_1,y_1)$ is **T = S$_1$**

where $T=\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1$, $S_1=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$, i.e. $\left(\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1\right)=\left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1\right)$

***Illustration 14 :*** Find the locus of the mid-point of focal chords of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

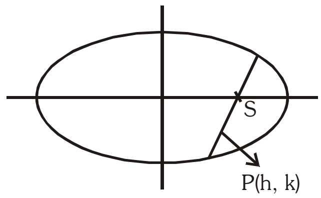
***Solution :*** Let P ≡ (h, k) be the mid-point

$\therefore$ equation of chord whose mid-point is given $\frac{xh}{a^2}+\frac{yk}{b^2}-1=\frac{h^2}{a^2}+\frac{k^2}{b^2}-1$

since it is a focal chord,

$\therefore$ It passes through focus, either (ae, 0) or (–ae, 0)

If it passes through (ae, 0)




$\therefore$ locus is $\frac{ex}{a}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$

If it passes through (–ae, 0)

$\therefore$ locus is $-\frac{ex}{a}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ **Ans.**

**Do yourself - 7 :**

**1.** Find the equation of chord of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ whose mid point be (–1, 1).

**2.** Find the locus of the middle points of chords of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

(a) whose distance from the centre is the constant length c.

(b) which subtend a right angle at the centre.

(c) which pass through the given point (h,k)

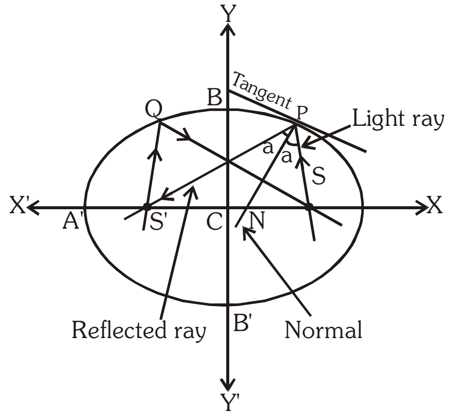
(d) whose length is constant (= 2c).

**3.** The angle between the pair of tangents drawn to the ellipse $2x^2+3y^2=6$ from the point (–1, 2), is

**4.** The equations of the tangents of the ellipse $3x^2+4y^2=12$ which pass through the point (2, 1) is

**5.** If the distance between directrices of an ellipse is 12 and its eccentricity is $\frac{1}{3}$, then find the radius of director circle of this ellipse.

## 14. IMPORTANT POINTS :




Referring to an ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$

**(a)** If P be any point on the ellipse with S & S' as its foci then $\ell(SP)+\ell(S'P)=2a$.

**(b)** The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice versa .

**(c)** **The product of the length's of the perpendicular segments from the foci on any tangent to the ellipse is $b^2$** and the feet of these perpendiculars lie on its auxiliary circle and the tangents at these feet to the auxiliary circle meet on the ordinate of P and that the locus of their point of intersection is a similar ellipse as that of the original one.

**(d)** The portion of the tangent to an ellipse between the point of contact & the directrix subtends a **right angle** at the corresponding focus.

**(e)** If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively, & if CF be perpendicular upon this normal, then

**(i)** $\mathbf{PF\ .\ PG = b^2}$ **(ii)** $\mathbf{PF\ .\ Pg = a^2}$

**(iii)** $\mathbf{PG\ .\ Pg = SP\ .\ S'P}$ **(iv)** $\mathbf{CG\ .\ CT = CS^2}$

**(v)** locus of the mid point of Gg is another ellipse having the same eccentricity as that of the original ellipse.

[where S and S' are the focii of the ellipse and T is the point where tangent at P meet the major axis]

**(f)** Atmost four normals & two tangents can be drawn from any point to an ellipse.

**(g)** The circle on any focal distance as diameter touches the auxiliary circle.

**(h)** Perpendiculars from the centre upon all chords which join the ends of any perpendicular diameters of the ellipse are of constant length.

**(i)** If the tangent at the point P of a standard ellipse meets the axes in T and t and CY is the perpendicular on it from the centre then,

**(i)** $\mathbf{Tt\ .\ PY = a^2 - b^2}$ **and** **(ii)** **least value of Tt is a + b.**

**Do yourself - 8 :**

**1.** A man running round a racecourse note that the sum of the distance of two flag-posts from him is always 20 meters and distance between the flag-posts is 16 meters. Find the area of the path be encloses in square meters

**2.** If chord of contact of the tangent drawn from the point $(\alpha, \beta)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ touches the circle $x^2 + y^2 = k^2$, then find the locus of the point $(\alpha, \beta)$.

**3.** The ratio of the area of triangle inscribed in an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ to that of triangle formed by the corresponding points on its auxiliary circle is 1/3 then, find the eccentricity of the ellipse.

**4.** If $F_1$ and $F_2$ are the feet of the perpendiculars from the foci $S_1$ and $S_2$, respectively, of an ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ on the tangent at any point P on the ellipse, then prove that $S_1F_1+S_2F_2\geq 8$.

**5.** The normal at point $P\left(\frac{\pi}{3}\right)$ on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ meets the major axis of the ellipse at Q. if S and S' are foci of given ellipse, then find the ratio SQ : S'Q.

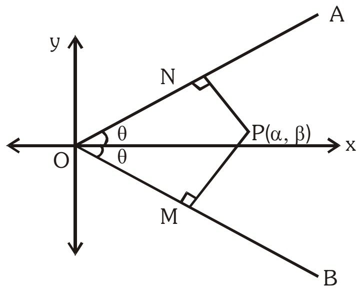
**6.** If focus and corresponding directrix of an ellipse are $(3, 4)$ and $x+y-1=0$ and eccentricity is $\frac{1}{2}$ then find the co-ordinates of extremities of major axis.

***Miscellaneous Illustration :***

***Illustration 15 :*** A point moves so that the sum of the squares of its distances from two intersecting straight lines is constant. Prove that its locus is an ellipse.

***Solution :*** Let two intersecting lines OA and OB, intersect at origin O and let both lines OA and OB makes equal angles with x axis.

i.e., $\angle XOA = \angle XOB = \theta$.




$\therefore$ Equations of straight lines OA and OB are

$y = x\tan\theta$ and $y = -x\tan\theta$

or $x\sin\theta - y\cos\theta = 0$ ......... (i)

and $x\sin\theta + y\cos\theta = 0$ ......... (ii)

Let $P(\alpha, \beta)$ is the point whose locus is to be determine.

According to the example $(PM)^2 + (PN)^2 = 2\lambda^2$ (say)

$\therefore (\alpha\sin\theta + \beta\cos\theta)^2 + (\alpha\sin\theta - \beta\cos\theta)^2 = 2\lambda^2 \Rightarrow 2\alpha^2\sin^2\theta + 2\beta^2\cos^2\theta = 2\lambda^2$

or $\alpha^2\sin^2\theta + \beta^2\cos^2\theta = \lambda^2 \Rightarrow \frac{\alpha^2}{\lambda^2\mathrm{cosec}^2\theta}+\frac{\beta^2}{\lambda^2\sec^2\theta}=1$

$$\Rightarrow \frac{\alpha^2}{(\lambda\ \mathrm{cosec}\ \theta)^2}+\frac{\beta^2}{(\lambda\sec\theta)^2}=1$$

Hence required locus is $\frac{x^2}{(\lambda\ \mathrm{cosec}\ \theta)^2}+\frac{y^2}{(\lambda\sec\theta)^2}=1$ **Ans.**

***Illustration 16 :*** Find the condition on 'a' (a > 0) and 'b' (b > 0) for which two distinct chords of the ellipse

$\frac{x^2}{2a^2}+\frac{y^2}{2b^2}=1$ passing through (a, –b) are bisected by the line x + y = b.

***Solution :*** Let (t, b – t) be a point on the line x + y = b.

Then equation of chord whose mid point (t, b – t) is

$$\frac{tx}{2a^2}+\frac{y(b-t)}{2b^2}-1=\frac{t^2}{2a^2}+\frac{(b-t)^2}{2b^2}-1 \qquad \text{.......... (i)}$$

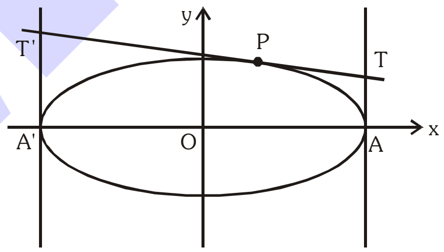
(a, –b) lies on (i) then $\frac{ta}{2a^2}-\frac{b(b-t)}{2b^2}=\frac{t^2}{2a^2}+\frac{(b-t)^2}{2b^2} \Rightarrow t^2(a^2+b^2)-ab(3a+b)t+2a^2b^2=0$

Since t is real and distinct $B^2 - 4AC > 0 \quad \Rightarrow \quad a^2b^2(3a+b)^2-4(a^2+b^2)2a^2b^2>0$

$\Rightarrow \quad a^2+6ab-7b^2>0 \qquad \Rightarrow \quad a^2+6ab>7b^2$, which is the required condition.

***Illustration 17 :*** Any tangent to an ellipse is cut by the tangents at the ends of the major axis in T and T '. Prove that circle on TT' as diameter passes through foci.

***Solution :*** Let ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$




and let P(acosϕ, bsinϕ) be any point on this ellipse

∴ Equation of tangent at P(acosϕ, bsinϕ) is

$$\frac{x}{a}\cos\phi+\frac{y}{b}\sin\phi=1 \qquad \text{.....(i)}$$

The two tangents drawn at the ends of the major axis are x = a and x = –a

Solving (i) and x = a we get $T=\left\{a,\frac{b(1-\cos\phi)}{\sin\phi}\right\}\equiv\left\{a, b\tan\left(\frac{\phi}{2}\right)\right\}$

and solving (i) and x = – a we get $T'=\left\{-a,\frac{b(1+\cos\phi)}{\sin\phi}\right\}\equiv\left\{-a, b\cot\left(\frac{\phi}{2}\right)\right\}$

Equation of circle on TT' as diameter is $(x-a)(x+a)+(y-b\tan(\phi/2))(y-b\cot(\phi/2))=0$

or $\quad x^2+y^2-by(\tan(\phi/2)+\cot(\phi/2))-a^2+b^2=0 \qquad$ ........ (ii)

Now put x = ± ae and y = 0 in LHS of (ii), we get

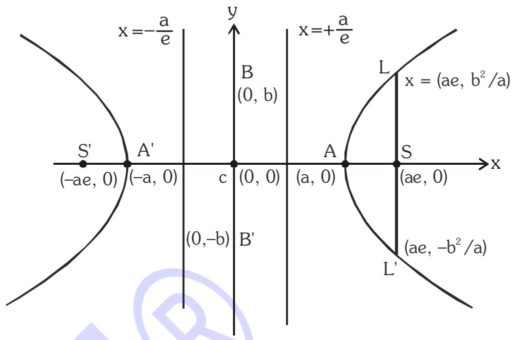
$a^2e^2+0-0-a^2+b^2=a^2-b^2-a^2+b^2=0=$ RHS

Hence foci lie on this circle

# HYPERBOLA

The **Hyperbola** is a conic whose eccentricity is greater than unity. **(e > 1)**.

## 1. STANDARD EQUATION & DEFINITION(S) :




Standard equation of the hyperbola is

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1, \text{ where } b^2=a^2(e^2-1)$$

or $a^2e^2=a^2+b^2$ i.e. $e^2=1+\frac{b^2}{a^2}$

$$=1+\left(\frac{\text{Conjugate Axis}}{\text{Transverse Axis}}\right)^2$$

**(a) Foci :**

$\mathbf{S \equiv (ae, 0)}$ & $\mathbf{S' \equiv (-ae, 0)}$.

**(b) Equations of directrices :**

$$x=\frac{a}{e} \quad \& \quad x=-\frac{a}{e}.$$

**(c) Vertices :**

$\mathbf{A \equiv (a, 0)}$ & $\mathbf{A' \equiv (-a, 0)}$.

**(d) Latus rectum :**

(i) Equation : $\mathbf{x = \pm ae}$

(ii) Length $=\frac{2b^2}{a}=\frac{(\text{Conjugate Axis})^2}{(\text{Transverse Axis})}=2a(e^2-1)=2e$ **(distance from focus to directrix)**

(iii) Ends : $\left(ae, \frac{b^2}{a}\right),\left(ae, \frac{-b^2}{a}\right);\left(-ae, \frac{b^2}{a}\right),\left(-ae, \frac{-b^2}{a}\right)$

**(e) (i) Transverse Axis :**

The line segment A′A of length 2a in which the foci S′ & S both lie is called the **Transverse Axis of the Hyperbola.**

**(ii) Conjugate Axis :**

The line segment B′B between the two points $B' \equiv (0, -b)$ & $B \equiv (0, b)$ is called as the **Conjugate Axis of the Hyperbola.**

The Transverse Axis & the Conjugate Axis of the hyperbola are together called the **Principal axes of the hyperbola.**

**(f) Focal Property :**

The difference of the focal distances of any point on the hyperbola is constant and equal to transverse axis i.e. $\left||PS| - |PS'|\right| = 2a$. The distance SS′ = focal length.

**(g) Focal distance :**

Distance of any point P(x, y) on Hyperbola from foci **PS = ex – a & PS′ = ex + a.**

---

***Illustration 1 :*** Find the equation of the hyperbola whose directrix is $2x + y = 1$, focus (1, 2) and eccentricity $\sqrt{3}$.

***Solution :*** Let P(x, y) be any point on the hyperbola and PM is perpendicular from P on the directrix.

Then by definition SP = e PM

$$\Rightarrow \quad (SP)^2 = e^2 (PM)^2 \quad \Rightarrow \quad (x-1)^2 + (y-2)^2 = 3\left\{\frac{2x+y-1}{\sqrt{4+1}}\right\}^2$$

$$\Rightarrow \quad 5(x^2 + y^2 - 2x - 4y + 5) = 3(4x^2 + y^2 + 1 + 4xy - 2y - 4x)$$

$$\Rightarrow \quad 7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$$

which is the required hyperbola.

***Illustration 2 :*** The eccentricity of the hyperbola $4x^2 - 9y^2 - 8x = 32$ is -

(A) $\frac{\sqrt{5}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{13}}{2}$ (D) $\frac{3}{2}$

***Solution :*** $4x^2 - 9y^2 - 8x = 32 \Rightarrow 4(x-1)^2 - 9y^2 = 36 \Rightarrow \frac{(x-1)^2}{9} - \frac{y^2}{4} = 1$

Here $a^2 = 9, b^2 = 4$

$$\therefore \quad \text{eccentricity } e = \sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+\frac{4}{9}} = \frac{\sqrt{13}}{3}$$ **Ans.(B)**

***Illustration 3 :*** If foci of a hyperbola are foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. If the eccentricity of the hyperbola be 2, then its equation is -

(A) $\frac{x^2}{4} - \frac{y^2}{12} = 1$ (B) $\frac{x^2}{12} - \frac{y^2}{4} = 1$ (C) $\frac{x^2}{12} + \frac{y^2}{4} = 1$ (D) none of these

***Solution :*** For ellipse $e = \frac{4}{5}$, so foci = (±4, 0)

For hyperbola e = 2, so $a = \frac{ae}{e} = \frac{4}{2} = 2,\ b = 2\sqrt{4-1} = 2\sqrt{3}$

Hence equation of the hyperbola is $\frac{x^2}{4} - \frac{y^2}{12} = 1$ **Ans.(A)**

***Illustration 4 :*** Find the coordinates of foci, the eccentricity and latus-rectum, equations of directrices for the hyperbola $9x^2 - 16y^2 - 72x + 96y - 144 = 0$.

***Solution :*** Equation can be rewritten as $\frac{(x-4)^2}{4^2} - \frac{(y-3)^2}{3^2} = 1$ so a = 4, b = 3

$b^2 = a^2(e^2 - 1)$ given $e = \frac{5}{4}$

Foci : $X = \pm ae$, $Y = 0$ gives the foci as (9, 3), (–1, 3)

Centre : $X = 0$, $Y = 0$ i.e. (4, 3)

Directrices : $X = \pm \frac{a}{e}$ i.e. $x - 4 = \pm \frac{16}{5}$ $\therefore$ directrices are $5x - 36 = 0$; $5x - 4 = 0$

Latus-rectum $= \frac{2b^2}{a} = 2.\frac{9}{4} = \frac{9}{2}$

## 2. CONJUGATE HYPERBOLA :

Two hyperbolas such that transverse & conjugate axes of one hyperbola are respectively the conjugate & the transverse axes of the other are called **Conjugate Hyperbolas** of each other. eg. $\mathbf{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$ & $\mathbf{-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}$ are conjugate hyperbolas of each other .

**Note that :**

(i) If $\mathbf{e_1}$ & $\mathbf{e_2}$ are the eccentricities of the hyperbola & its conjugate then $\mathbf{e_1^{-2} + e_2^{-2} = 1}$.

(ii) The foci of a **hyperbola** and its **conjugate** are **concyclic and form the vertices of a square.**

(iii) Two hyperbolas are said to be **similar** if they have the **same eccentricity**.

***Illustration 5 :*** The eccentricity of the conjugate hyperbola to the hyperbola $x^2 - 3y^2 = 1$ is-

(A) 2 (B) $2/\sqrt{3}$ (C) 4 (D) 4/3

***Solution :*** Equation of the conjugate hyperbola to the hyperbola $x^2 - 3y^2 = 1$ is

$-x^2 + 3y^2 = 1 \quad \Rightarrow \quad -\frac{x^2}{1} + \frac{y^2}{1/3} = 1$

Here $a^2 = 1$, $b^2 = 1/3$

$\therefore$ eccentricity $e = \sqrt{1 + a^2/b^2} = \sqrt{1+3} = 2$ **Ans. (A)**

**Do yourself - 1 :**

**1.** Find the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ which passes through (4, 0) & $(3\sqrt{2}, 2)$

**2.** Find the equation to the hyperbola, whose eccentricity is $\frac{5}{4}$, focus is (a, 0) and whose directrix is $4x - 3y = a$.

**3.** In the hyperbola $4x^2 - 9y^2 = 36$, find length of the axes, the co-ordinates of the foci, the eccentricity, and the latus rectum.

**4.** Find the equation to the hyperbola, the distance between whose foci is 16 and whose eccentricity is $\sqrt{2}$.

**5.** Consider the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$. Find the following:

**(a)** centre **(b)** eccentricity **(c)** focii **(d)** equation of directrix

**(e)** length of the latus rectum

**6.** Find the equation to the hyperbola, referred to its axes as axes of coordinates,

(a) whose transverse and conjugate axes are respectively 3 and 4,

(b) whose conjugate axis is 5 and the distance between whose foci is 13

(c) whose conjugate axis is 7 and which passes through the point (3,–2)

**7.** Find the equation to the hyperbola of given transverse axis whose vertex bisects the distance between the centre and the focus.

**8.** Find the equation to the hyperbola, whose eccentricity is $\frac{5}{4}$, whose focus is (a,0), and whose directrix is $4x - 3y = a$.

Find also the coordinates of the centre and the equation to the other directrix.

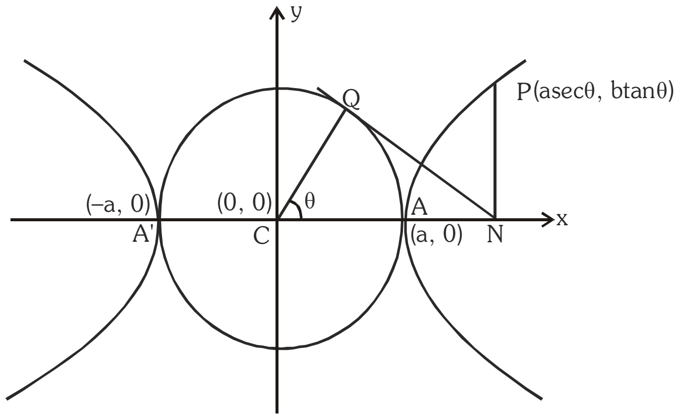
**9.** Find eccentricity of conjugate hyperbola of hyperbola $4x^2 - 16y^2 = 64$, also find area of quadrilateral formed by foci of hyperbola & its conjugate hyperbola

## 3. RECTANGULAR OR EQUILATERAL HYPERBOLA :

The particular kind of hyperbola in which the lengths of the transverse & conjugate axis are equal is called an **Equilateral Hyperbola**. Note that the eccentricity of the rectangular hyperbola is $\sqrt{2}$ and **the length of it's latus rectum is equal to it's transverse or conjugate axis.**

## 4. AUXILIARY CIRCLE :

A circle drawn with centre C & T.A. as a diameter is called the **Auxiliary Circle** of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$ .




Note from the figure that P & Q are called the **"Corresponding Points"** on the hyperbola & the auxiliary circle. 'θ' is called the **eccentric angle** of the point 'P' on the hyperbola. $(0 \le \theta < 2\pi)$.

**Parametric Equation :**

The equations $x = a \sec\theta$ & $y = b \tan\theta$ together represents the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where θ is a parameter. The parametric equations ; $x = a \cosh\phi, y = b \sinh\phi$ also represents the same hyperbola.

**General Note :**

Since the fundamental equation to the hyperbola only differs from that to the ellipse in having $-b^2$ instead of $b^2$ it will be found that many propositions for the hyperbola are derived from those for the ellipse by simply changing the sign of $b^2$ .

## 5. POSITION OF A POINT 'P' w.r.t. A HYPERBOLA :

The quantity $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$ is **positive , zero** or **negative** according as the point $(x_1, y_1)$ lies within , upon or outside the curve.

## 6. LINE AND A HYPERBOLA :

The straight line $y = mx + c$ is a secant , a tangent or passes outside the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ according as : $c^2 > = < a^2m^2 - b^2$.

Equation of a chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ joining its two points **P(α) & Q(β)** is

$$\frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}$$

***Illustration 6 :*** Show that the line x cos α + y sin α = p touches the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if $a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$.

***Solution :*** The given line is $x\cos\alpha + y\sin\alpha = p \Rightarrow y\sin\alpha = -x\cos\alpha + p$

$\Rightarrow \quad y = -x\cot\alpha + p\,\text{cosec}\,\alpha$

Comparing this line with y = mx + c

$m = -\cot\alpha,\ c = p\,\text{cosec}\,\alpha$

Since the given line touches the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ then

$c^2 = a^2m^2 - b^2 \quad \Rightarrow \quad p^2\,\text{cosec}^2\alpha = a^2\cot^2\alpha - b^2$ or $p^2 = a^2\cos^2\alpha - b^2\sin^2\alpha$

***Illustration 7 :*** If (a secθ, b tanθ) and (asecϕ, b tanϕ) are the ends of a focal chord of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then $\tan\frac{\theta}{2}\tan\frac{\phi}{2}$ equal to -

(A) $\frac{e-1}{e+1}$ (B) $\frac{1-e}{1+e}$ (C) $\frac{1+e}{1-e}$ (D) $\frac{e+1}{e-1}$

***Solution :*** Equation of chord connecting the points (asecθ, b tanθ) and (asecϕ, b tanϕ) is

$$\frac{x}{a}\cos\left(\frac{\theta-\phi}{2}\right)-\frac{y}{b}\sin\left(\frac{\theta+\phi}{2}\right)=\cos\left(\frac{\theta+\phi}{2}\right) \qquad \text{........ (i)}$$

If it passes through (ae, 0); we have, $e\cos\left(\frac{\theta-\phi}{2}\right)=\cos\left(\frac{\theta+\phi}{2}\right)$

$$\Rightarrow \quad e=\frac{\cos\left(\frac{\theta+\phi}{2}\right)}{\cos\left(\frac{\theta-\phi}{2}\right)}=\frac{1-\tan\frac{\theta}{2}.\tan\frac{\phi}{2}}{1+\tan\frac{\theta}{2}.\tan\frac{\phi}{2}} \Rightarrow \quad \tan\frac{\theta}{2}.\tan\frac{\phi}{2}=\frac{1-e}{1+e}$$

Similarly if (i) passes through (–ae, 0), $\tan\frac{\theta}{2}.\tan\frac{\phi}{2}=\frac{1+e}{1-e}$ **Ans. (B, C)**

**Do yourself - 2 :**

**1.** Find the condition for the line $\ell x + my + n = 0$ to touch the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

**2.** If the line $y = 5x + 1$ touch the hyperbola $\frac{x^2}{4}-\frac{y^2}{b^2}=1$ $\{b > 4\}$, then -

(A) $b^2=\frac{1}{5}$ (B) $b^2 = 99$ (C) $b^2 = 4$ (D) $b^2 = 100$

**3.** Find the points common to the hyperbola $25x^2 - 9y^2 = 225$ and the straight line $25x + 12y - 45 = 0$.

**4.** Prove that the straight lines $\frac{x}{a}-\frac{y}{b}=m$ and $\frac{x}{a}+\frac{y}{b}=\frac{1}{m}$ always meet on the hyperbola.

**5.** For what value of $\lambda$, does the line $y = 3x + \lambda$ touch the hyperbola $9x^2 - 5y^2 = 45$ ?

**6.** If the straight line $2x+\sqrt{2}y+n=0$ touches the hperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$, then find the value of n.

## 7. TANGENT TO THE HYPERBOLA $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ :

**(a)** **Point form :** Equation of the tangent to the given hyperbola at the point $(x_1, y_1)$ is $\mathbf{\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1}$.

**Note :** In general two tangents can be drawn from an external point $(x_1 y_1)$ to the hyperbola and they are $\mathbf{y - y_1 = m_1 (x - x_1)}$ & $\mathbf{y - y_1 = m_2 (x - x_1)}$, where $\mathbf{m_1}$ & $\mathbf{m_2}$ are roots of the equation $\mathbf{(x_1^2 - a^2) m^2 - 2 x_1y_1 m + y_1^2 + b^2 = 0}$. If $\mathbf{D < 0}$, then **no tangent** can be drawn from $(x_1 y_1)$ to **the hyperbola.**

**(b)** **Slope form :** The equation of tangents of slope m to the given hyperbola is $\mathbf{y = m x \pm\sqrt{a^2m^2-b^2}}$. Point of contact are $\left(\mp\frac{\mathbf{a^2m}}{\sqrt{\mathbf{a^2m^2-b^2}}},\frac{\mathbf{\mp b^2}}{\sqrt{\mathbf{a^2m^2-b^2}}}\right)$

Note that there **are two parallel tangents having the same slope m.**

**(c)** **Parametric form :** Equation of the tangent to the given hyperbola at the point $(\mathbf{a\sec\theta, b\tan\theta})$ is $\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$.

**Note :** Point of intersection of the tangents at $\theta_1$ & $\theta_2$ is $\mathbf{x = a}\frac{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}{\cos\left(\frac{\theta_1+\theta_2}{2}\right)}$, $\mathbf{y = b\tan}\left(\frac{\theta_1+\theta_2}{2}\right)$

***Illustration 8 :*** Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.

***Solution :*** Let m be the slope of the tangent. Since the tangent is perpendicular to the line $x - y = 0$

$\therefore \quad m \times 1 = -1 \Rightarrow m = -1$

Since $x^2 - 4y^2 = 36$ or $\frac{x^2}{36} - \frac{y^2}{9} = 1$

Comparing this with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\therefore \quad a^2 = 36$ and $b^2 = 9$

So the equation of tangents are $y = (-1)x \pm \sqrt{36 \times (-1)^2 - 9}$

$y = -x \pm \sqrt{27} \quad \Rightarrow \quad x + y \pm 3\sqrt{3} = 0$ **Ans.**

***Illustration 9 :*** The locus of the point of intersection of two tangents of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if the product of their slopes is $c^2$, will be -

(A) $y^2 - b^2 = c^2(x^2 + a^2)$ (B) $y^2 + b^2 = c^2(x^2 - a^2)$

(C) $y^2 + a^2 = c^2(x^2 - b^2)$ (D) $y^2 - a^2 = c^2(x^2 + b^2)$

***Solution :*** Equation of any tangent of the hyperbola with slope m is $y = mx \pm \sqrt{a^2m^2 - b^2}$

If it passes through $(x_1, y_1)$ then

$(y_1 - mx_1)^2 = a^2m^2 - b^2 \quad \Rightarrow \quad (x_1^2 - a^2)\, m^2 - 2x_1y_1m + (y_1^2 + b^2) = 0$

If $m = m_1, m_2$ then as given $m_1m_2 = c^2 \quad \Rightarrow \quad \frac{y_1^2 + b^2}{x_1^2 - a^2} = c^2$

Hence required locus will be : $y^2 + b^2 = c^2(x^2 - a^2)$ **Ans.(B)**

***Illustration 10 :*** A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is -

(A) $y = 3\sqrt{\frac{2}{7}}x - \frac{15}{\sqrt{7}}$ (B) $y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$ (C) $y = -3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$ (D) $y = -3\sqrt{\frac{2}{7}}x - \frac{15}{\sqrt{7}}$

***Solution :*** $\frac{x^2}{16} - \frac{y^2}{9} = 1,\ x^2 + y^2 = 9$

Equation of tangent $y = mx + \sqrt{16m^2 - 9}$ (for hyperbola)

Equation of tangent $y = m'x + 3\sqrt{1 + m'^2}$ (circle)

For common tangent $m = m'$ and $3\sqrt{1+m'^2} = \sqrt{16m^2-9}$

or $\quad 9 + 9m^2 = 16m^2 - 9$

or $\quad 7m^2 = 18 \quad \Rightarrow \quad m = \pm 3\sqrt{\frac{2}{7}}$

$\therefore$ required equation is $y = \pm 3\sqrt{\frac{2}{7}}\, x \pm 3\sqrt{1+\frac{18}{7}}$

or $\quad y = \pm 3\sqrt{\frac{2}{7}}\, x \pm \frac{15}{\sqrt{7}}$ **Ans. (A,B,C,D)**

**Do yourself - 3 :**

**1.** Find the equation of the tangent to the hyperbola $4x^2 - 9y^2 = 1$, which is parallel to the line $4y = 5x + 7$.

**2.** Find the equation of the tangent to the hyperbola $16x^2 - 9y^2 = 144$ at $\left(5, \frac{16}{3}\right)$.

**3.** Find the common tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and an ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$.

**4.** Consider the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$. Find the following:
**(a)** equation of auxilary circle **(b)** equation of director circle

**5.** Show that the line $x\cos\alpha + y\sin\alpha = p$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$.

## 8. NORMAL TO THE HYPERBOLA $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ :

**(a)** **Point form :** The equation of the normal to the given hyperbola at the point $P(x_1, y_1)$ on it is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 = a^2e^2$.

**(b)** **Slope form :** The equation of normal of slope m to the given hyperbola is $y = mx \mp \frac{m(a^2+b^2)}{\sqrt{(a^2-m^2b^2)}}$ foot of normal are $\left(\pm\frac{a^2}{\sqrt{(a^2-m^2b^2)}}, \mp\frac{mb^2}{\sqrt{(a^2-m^2b^2)}}\right)$

**(c)** **Parametric form :** The equation of the normal at the point **P (a secθ , b tanθ)** to the given hyperbola is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2 = a^2e^2$.

***Illustration 11 :*** Line $x \cos \alpha + y \sin \alpha = p$ is a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if -

(A) $a^2 \sec^2\alpha - b^2 \operatorname{cosec}^2\alpha = \frac{(a^2+b^2)^2}{p^2}$ (C) $a^2 \sec^2\alpha + b^2 \operatorname{cosec}^2\alpha = \frac{(a^2+b^2)^2}{p^2}$

(C) $a^2 \cos^2\alpha - b^2 \sin^2\alpha = \frac{(a^2+b^2)^2}{p^2}$ (D) $a^2 \cos^2\alpha + b^2 \sin^2\alpha = \frac{(a^2+b^2)^2}{p^2}$

***Solution :*** Equation of a normal to the hyperbola is $ax \cos\theta + by \cot\theta = a^2 + b^2$

comparing it with the given line equation

$$\frac{a\cos\theta}{\cos\alpha} = \frac{b\cot\theta}{\sin\alpha} = \frac{a^2+b^2}{p} \quad \Rightarrow \quad \sec\theta = \frac{ap}{\cos\alpha(a^2+b^2)},\ \tan\theta = \frac{bp}{\sin\alpha(a^2+b^2)}$$

Eliminating $\theta$, we get

$$\frac{a^2p^2}{\cos^2\alpha(a^2+b^2)^2} - \frac{b^2p^2}{\sin^2\alpha(a^2+b^2)^2} = 1 \quad \Rightarrow \quad a^2\sec^2\alpha - b^2\operatorname{cosec}^2\alpha = \frac{(a^2+b^2)^2}{p^2}$$

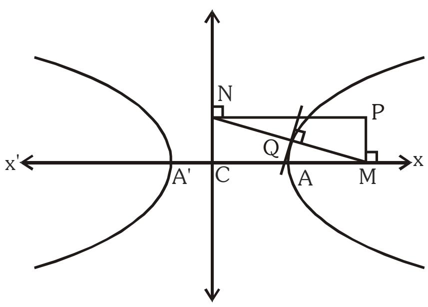
**Ans.(A)**

***Illustration 12 :*** The normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the axes in M and N, and lines MP and NP are drawn at right angles to the axes. Prove that the locus of P is hyperbola $(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$.

***Solution :*** Equation of normal at any point Q is $ax\cos\theta + by\cot\theta = a^2 + b^2$

$$\therefore \quad M \equiv \left(\frac{a^2+b^2}{a}\sec\theta,\ 0\right),\ N \equiv \left(0,\ \frac{a^2+b^2}{b}\tan\theta\right)$$

$\therefore$ Let $P \equiv (h, k)$




$$\Rightarrow \quad h = \frac{a^2+b^2}{a}\sec\theta, \qquad k = \frac{a^2+b^2}{b}\tan\theta$$

$$\Rightarrow \quad \frac{a^2h^2}{(a^2+b^2)} - \frac{b^2k^2}{(a^2+b^2)^2} = \sec^2\theta - \tan^2\theta = 1$$

$\therefore$ locus of P is $(a^2x^2 - b^2y^2) = (a^2 + b^2)$.

**Do yourself - 4 :**

**1.** Find the equation of normal to the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ at (5, 0).

**2.** Find the equation of normal to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ at the point $\left(6, \frac{3}{2}\sqrt{5}\right)$.

**3.** Find the condition for the line $\ell x + my + n = 0$ is normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

**4.** Find the equation of normal to the hyperbola $3x^2 - 4y^2 = 12$ having slope 1.

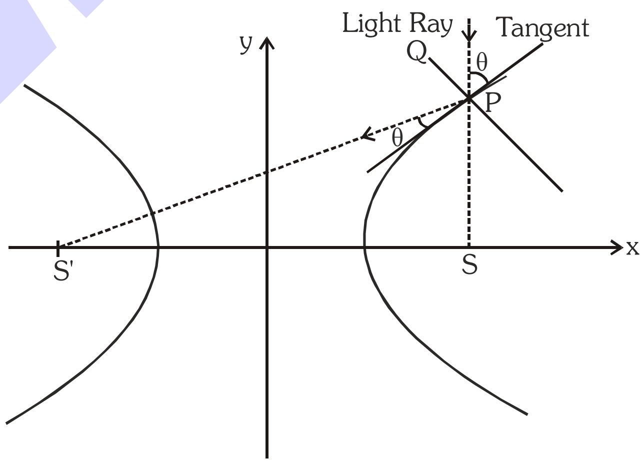
**5.** If a chord joining the points $P(a\sec\theta, a\tan\theta)$ & $Q(a\sec\phi, a\tan\phi)$ on the hyperbola $x^2 - y^2 = a^2$ is a normal to it at P, then show that $\tan\phi = \tan\theta(4\sec^2\theta - 1)$.

## 9. HIGHLIGHTS ON TANGENT AND NORMAL :

**(a)** Locus of the feet of the perpendicular drawn from focus of the hyperbola $\mathbf{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$ upon any tangent is its auxiliary circle i.e. $\mathbf{x^2 + y^2 = a^2}$ & the product of lengths to these perpendiculars is $\mathbf{b^2}$ **(semi Conjugate Axis)$^2$**

**(b)** The portion of the tangent between the **point of contact** & **the directrix** subtends a **right angle** at the corresponding **focus**.

**(c)** The tangent & normal at any point of a hyperbola **bisect** the angle between the **focal radii**. This spells the **reflection property of the hyperbola** as **"An incoming light ray"** aimed towards one **focus is reflected from the outer surface of the hyperbola towards the other focus.** It follows that if an ellipse and a hyperbola have the same foci, they cut at **right angles** at any of their **common point.**




**Note that the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ & the hyperbola $\frac{x^2}{a^2-k^2}-\frac{y^2}{k^2-b^2}=1$ $(a > k > b > 0)$ are confocal and therefore orthogonal.**

**(d)** The **foci of the hyperbola** and the **points P** and **Q** in which any tangent meets the tangents at the vertices are **concyclic** with **PQ** as **diameter of the circle.**

## 10. DIRECTOR CIRCLE :

The locus of the intersection of tangents which are at **right angles** is known as the **Director Circle** of the hyperbola. The equation to the **director circle is** : $x^2 + y^2 = a^2 - b^2$.

If $b^2 < a^2$, this circle is **real ;** if $b^2 = a^2$ the **radius** of the **circle is zero** & it reduces to a **point at the origin.** In this case tangent, coincides with asymptotes so no tangents at right angle can be drawn to the curve.

If $b^2 > a^2$, the **radius** of the **circle is imaginary**, so that there is **no such circle** & so **no tangents at right angle** can be drawn to the curve.

**Note :** Equations of chord of contact, chord with a given middle point, pair of tangents from an external point are to be interpreted in the similar way as in ellipse.

## 11. ASYMPTOTES :

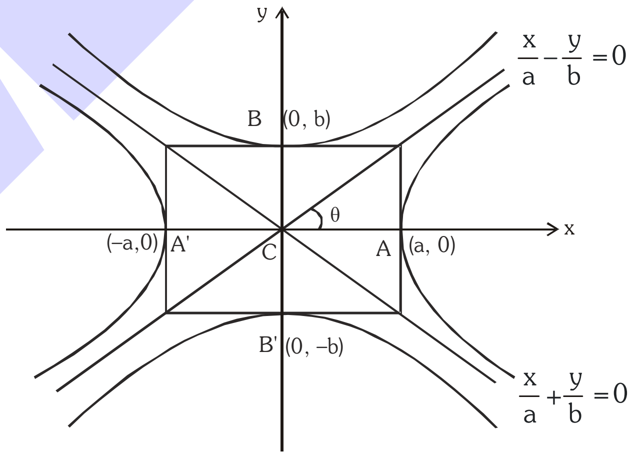
**Definition :** If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called the **Asymptote of the Hyperbola**.

**To find the asymptote of the hyperbola :**

Let $y = mx + c$ is the **asymptote** of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ .

Solving these two we get the quadratic as $(b^2 - a^2m^2)\,x^2 - 2a^2mcx - a^2(b^2 + c^2) = 0$ .........(1)

In order that $y = mx + c$ **be an asymptote**, both **roots** of equation (1) must approach infinity, the conditions for which are : **coefficient of $x^2 = 0$ & coefficient of $x = 0$.**




$\Rightarrow$ $b^2 - a^2m^2 = 0$ or $m = \pm\frac{b}{a}$ &

$a^2mc = 0 \Rightarrow c = 0$.

$\therefore$ equations of asymptote are $\frac{x}{a} + \frac{y}{b} = 0$

and $\frac{x}{a} - \frac{y}{b} = 0$.

combined equation to the asymptotes $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ .

**Particular Case :**

When $b = a$ the asymptotes of the rectangular hyperbola.

$x^2 - y^2 = a^2$ are $y = \pm x$ which are at **right angles**.

**Note :**

(i) **Equilateral hyperbola $\Leftrightarrow$ rectangular hyperbola.**

(ii) If a **hyperbola is equilateral** then the **conjugate hyperbola is also equilateral**.

(iii) A **hyperbola** and its **conjugate** have the **same asymptote.**

(iv) The equation of the **pair of asymptotes** differ the **hyperbola** & **the conjugate hyperbola** by the **same constant** only.

(v) The asymptotes pass through the **centre of the hyperbola** & the bisectors of the angles between the asymptotes are the axes of the hyperbola.

(vi) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.

(vii) Asymptotes are not tangent to the hyperbola as there is no common point between asymptotes and hyperbola.

(viii) A simple method to find the co–ordinates of the centre of the hyperbola expressed as a general equation of degree 2 should be remembered as : Let $\mathbf{f(x,y)=0}$ represents a hyperbola.

Find $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}$ & $\frac{\partial \mathbf{f}}{\partial \mathbf{y}}$. Then the point of intersection of $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\mathbf{=0}$ & $\frac{\partial \mathbf{f}}{\partial \mathbf{y}}\mathbf{=0}$ gives the **centre** of the **hyperbola**.

***Illustration 13 :*** Find the asymptotes of the hyperbola $2x^2 + 5xy + 2y^2 + 4x + 5y = 0$. Find also the general equation of all the hyperbolas having the same set of asymptotes.

***Solution :*** Let $2x^2 + 5xy + 2y^2 + 4x + 5y + \lambda = 0$ be asymptotes. This will represent two straight line so

$$4\lambda + 25 - \frac{25}{2} - 8 - \frac{25}{4}\lambda = 0$$

$\Rightarrow$ $\lambda = 2$

$\Rightarrow$ $2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0$ are asymptotes

$\Rightarrow$ $(2x + y + 2) = 0$ and $(x + 2y + 1) = 0$ are asymptotes

and $2x^2 + 5xy + 2y^2 + 4x + 5y + c = 0$ is general equation of hyperbola $\forall\ c \in \mathbb{R}\backslash\{2\}$.

***Illustration 14 :*** Find the hyperbola whose asymptotes are $2x - y = 3$ and $3x + y - 7 = 0$ and which passes through the point (1, 1).

***Solution :*** The equation of the hyperbola differs from the equation of the asymptotes by a constant

$\Rightarrow$ The equation of the hyperbola with asymptotes $3x + y - 7 = 0$ and $2x - y = 3$ is $(3x + y - 7)(2x - y - 3) + k = 0$

It passes through (1, 1)

$\Rightarrow$ $k = -6$.

Hence the equation of the hyperbola is $(2x - y - 3)(3x + y - 7) = 6$.

**Do yourself - 5 :**

**1.** Find the equation to the chords of the hyperbola $x^2 - y^2 = 9$ which is bisected at (5, –3)

**2.** If $m_1$ and $m_2$ are the slopes of the tangents to the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ which pass through the point (6, 2), then find the value of $11m_1m_2$ and $11(m_1 + m_2)$.

**3.** Find the locus of the mid points of the chords of the circle $x^2 + y^2 = 16$ which are tangents to the hyperbola $9x^2 - 16y^2 = 144$.

**4.** The asymptotes of a hyperbola are parallel to lines $2x + 3y = 0$ and $3x + 2y = 0$. The hyperbola has its centre at (1, 2) and it passes through (5, 3). Find its equation.

**5.** Find the equation of hyperbola having foci S(2, 1) and S(10, 1) and a straight line $x + y - 9 = 0$ as its tangent. Also, find the equation of its director circle.

**6.** A ray emanating from the point (5, 0) is incident on the hyperbola $9x^2 - 16y^2 = 144$ at the point $P(8, 3\sqrt{3})$. Find the equation of the reflected ray after first reflection.

## 12. HIGHLIGHTS ON ASYMPTOTES

**(a)** If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.

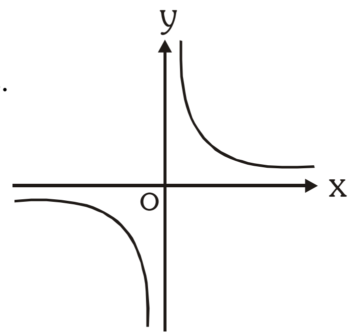
**(b)** Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.

**(c)** The tangent at any point P on a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with **centre C**, meets the **asymptotes in Q** and **R** and cuts off a **Δ CQR** of **constant area** equal to **ab** from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact . This implies that locus of the centre of the circle circumscribing the **Δ CQR** in case of a rectangular hyperbola is the **hyperbola** itself.

**(d)** If the angle between the asymptote of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is **2θ** then the eccentricity of the **hyperbola** is **sec θ**.

## 13. RECTANGULAR HYPERBOLA :

Rectangular hyperbola referred to its asymptotes as axis of coordinates.




**(a)** Equation is **xy = c²** with parametric representation

$x = ct, \quad y = c/t, \quad t \in R - \{0\}$.

**(b)** Equation of a chord joining the points **P(t₁) & Q(t₂)** is

$x + t_1t_2\,y = c(t_1 + t_2)$ with slope, $m = \frac{-1}{t_1t_2}$

**(c)** Equation of the tangent at $\mathbf{P(x_1, y_1)}$ is $\mathbf{\frac{x}{x_1} + \frac{y}{y_1} = 2}$

& at P(t) is $\mathbf{\frac{x}{t} + t\,y = 2\,c.}$

**(d)** Equation of normal is $\mathbf{y - \frac{c}{t} = t^2(x - ct)}$

**(e)** Chord with a given middle point as **(h, k)** is **kx + hy = 2hk.**

**Note :**

For the hyperbola, $\mathbf{xy = c^2}$

(i) Vertices : **(c, c) & (–c, –c)**.

(ii) Foci : $\mathbf{(\sqrt{2}c, \sqrt{2}c)}$ & $\mathbf{(-\sqrt{2}c, -\sqrt{2}c)}$

(iii) Directrices : $\mathbf{x + y = \pm\sqrt{2}c}$

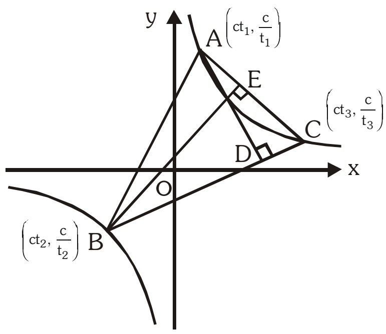
(iv) Latus rectum : $\mathbf{\ell = 2\sqrt{2}c = T\,.\,A = C\,.\,A}$

***Illustration 15 :*** A triangle has its vertices on a rectangular hyperbola. Prove that the orthocentre of the triangle also lies on the same hyperbola.

***Solution :*** Let $t_1$, $t_2$ and $t_3$ are the vertices of the triangle ABC, described on the rectangular hyperbola $xy = c^2$.

$\therefore$ co-ordinates of A, B and C are $\left(ct_1, \frac{c}{t_1}\right)$, $\left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ respectively

Now slope of BC is $\dfrac{\frac{c}{t_3} - \frac{c}{t_2}}{ct_3 - ct_2} = -\dfrac{1}{t_2t_3}$




$\therefore$ Slope of AD is $t_2t_3$

Equation of altitude AD is $y - \frac{c}{t_1} = t_2t_3(x - ct_1)$

or $t_1y - c = xt_1t_2t_3 - ct_1^2t_2t_3$ ........ (i)

Similarly equation of altitude BE is

$t_2y - c = xt_1t_2t_3 - ct_1t_2^2t_3$ ........ (ii)

Solving (i) and (ii), we get the orthocentre $\left(-\frac{c}{t_1t_2t_3}, -ct_1t_2t_3\right)$ which lies on $xy = c^2$.

**Do yourself - 6 :**

**1.** If equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola then write required conditions.

**2.** Find the equation of tangent at the point (1, 2) to the rectangular hyperbola $xy = 2$.

**3.** Prove that the locus of point, tangents from where to hyperbola $x^2 - y^2 = a^2$ inclined at an angle $\alpha$ & $\beta$ with x-axis such that $\tan\alpha \tan\beta = 2$ is also a hyperbola. Find the eccentricity of this hyperbola.

**4.** Find everything for the conic $xy = -36$.

**5.** Consider hyperbola $xy = 16$ to find the following:

(i) Equation of tangent at point (2, 8)

(ii) Equation of chord of contact w.r.t. point (2, 3)

(iii) Equation of chord which gets bisected at point (5, 6)

(iv) Equation of normal having slope 2.

***Miscellaneous Illustrations :***

***Illustration 16 :*** Chords of the circle $x^2 + y^2 = a^2$ touch the hyperbola $x^2/a^2 - y^2/b^2 = 1$. Prove that locus of their middle point is the curve $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$.

***Solution :*** Let (h, k) be the mid-point of the chord of the circle $x^2 + y^2 = a^2$, so that its equation by $T = S_1$ is $hx + ky = h^2 + k^2$

or $y = -\frac{h}{k}x + \frac{h^2+k^2}{k}$ i.e. of the form $y = mx + c$

It will touch the hyperbola if $c^2 = a^2m^2 - b^2$

$$\therefore \left(\frac{h^2+k^2}{k}\right)^2 = a^2\left(-\frac{h}{k}\right)^2 - b^2 \quad \text{or} \quad (h^2 + k^2)^2 = a^2h^2 - b^2k^2$$

Generalising, the locus of mid-point (h, k) is $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$

***Illustration 17 :*** C is the centre of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The tangent at any point P on this hyperbola meets the straight lines $bx - ay = 0$ and $bx + ay = 0$ in the points Q and R respectively. Show that $CQ \cdot CR = a^2 + b^2$.

***Solution :*** P is $(a\sec\theta, b\tan\theta)$

Tangent at P is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$

It meets $bx - ay = 0$ i.e. $\frac{x}{a} = \frac{y}{b}$ in Q

$\therefore$ Q is $\left(\dfrac{a}{\sec\theta-\tan\theta}, \dfrac{b}{\sec\theta-\tan\theta}\right)$

It meets bx + ay = 0 i.e. $\dfrac{x}{a}=-\dfrac{y}{b}$ in R.

$\therefore$ R is $\left(\dfrac{a}{\sec\theta+\tan\theta}, \dfrac{-b}{\sec\theta+\tan\theta}\right)$

$\therefore$ $CQ.CR = \dfrac{\sqrt{(a^2+b^2)}}{\sec\theta-\tan\theta}.\dfrac{\sqrt{(a^2+b^2)}}{\sec\theta+\tan\theta} = a^2 + b^2$ $(\because \quad \sec^2\theta - \tan^2\theta = 1)$

**Ans.**

***Illustration 18 :*** A circle with fix centre (h,k) and of variable radius cuts the rectangular hyperbola $x^2 - y^2 = 9a^2$ in points P, Q, R and S. Determine the equation of the locus of the centroid of triangle PQR.

***Solution :*** Let the circle be $(x - h)^2 + (y - k)^2 = r^2$ where r is variable. Its intersection with $x^2 - y^2 = 9a^2$ is obtained by putting $y^2 = x^2 - 9a^2$.

$x^2 + x^2 - 9a^2 - 2hx + h^2 + k^2 - r^2 = 2k\sqrt{(x^2-9a^2)}$

or $[2x^2 - 2hx + (h^2 + k^2 - r^2 - 9a^2)]^2 = 4k^2(x^2 - 9a^2)$

or $4x^4 - 8hx^3 + ..... = 0$

$\therefore$ Above gives the abscissas of the four points of intersection.

$\therefore$ $\Sigma x_1 = \dfrac{8h}{4} = 2h$

$x_1 + x_2 + x_3 + x_4 = 2h$

Similarly $y_1 + y_2 + y_3 + y_4 = 2k$.

Now if $(\alpha, \beta)$ be the centroid of $\Delta PQR$, then $3\alpha = x_1 + x_2 + x_3$, $3\beta = y_1 + y_2 + y_3$

$\therefore$ $x_4 = 2h - 3\alpha$, $y_4 = 2k - 3\beta$

But $(x_4, y_4)$ lies on $x^2 - y^2 = 9a^2$

$\therefore$ $(2h - 3\alpha)^2 - (2k - 3\beta)^2 = 9a^2$

Hence the locus of centroid $(\alpha, \beta)$ is $(2h - 3x)^2 - (2k - 3y)^2 = 9a^2$

or $\left(x-\dfrac{2h}{3}\right)^2-\left(y-\dfrac{2k}{3}\right)^2 = a^2$

***Illustration 19 :*** If a circle cuts a rectangular hyperbola $xy = c^2$ in A, B, C, D and the parameters of these four points be $t_1$, $t_2$, $t_3$ and $t_4$ respectively, then prove that :

(a) $t_1t_2t_3t_4 = 1$

(b) The centre of mean position of the four points bisects the distance between the centres of the two curves.

***Solution :*** (a) Let the equation of the hyperbola referred to rectangular asymptotes as axes be $xy = c^2$ or its parametric equation be

$x = ct,\ y = c/t$ .......... (i)

and that of the circle be

$x^2 + y^2 + 2gx + 2fy + k = 0$ .......... (ii)

Solving (i) and (ii), we get

$$c^2t^2 + \frac{c^2}{t^2} + 2gct + 2f\frac{c}{t} + k = 0$$

or $\quad c^2t^4 + 2gct^3 + kt^2 + 2fct + c^2 = 0$ .......... (iii)

Above equation being of fourth degree in t gives us the four parameters $t_1$, $t_2$, $t_3$, $t_4$ of the points of intersection.

$$\therefore \quad t_1 + t_2 + t_3 + t_4 = -\frac{2gc}{c^2} = -\frac{2g}{c} \qquad \text{.......... (iv)}$$

$$t_1t_2t_3 + t_1t_2t_4 + t_3t_4t_1 + t_3t_4t_2$$

$$= -\frac{2fc}{c^2} = -\frac{2f}{c} \qquad \text{.......... (v)}$$

$$t_1t_2t_3t_4 = \frac{c^2}{c^2} = 1. \text{ It proves (a)} \qquad \text{.......... (vi)}$$

Dividing (v) by (vi), we get

$$\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4} = -\frac{2f}{c} \qquad \text{.......... (vii)}$$

(b) The centre of mean position of the four points of intersection is

$$\left[\frac{c}{4}(t_1 + t_2 + t_3 + t_4),\ \frac{c}{4}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4}\right)\right] = \left[\frac{c}{4}\left(-\frac{2g}{c}\right),\ \frac{c}{4}\left(-\frac{2f}{c}\right)\right], \text{ by (iv) and (vii)}$$

$= (-g/2, -f/2)$

Above is clearly the mid-point of (0, 0) and $(-g, -f)$ i.e. the join of the centres of the two curves.